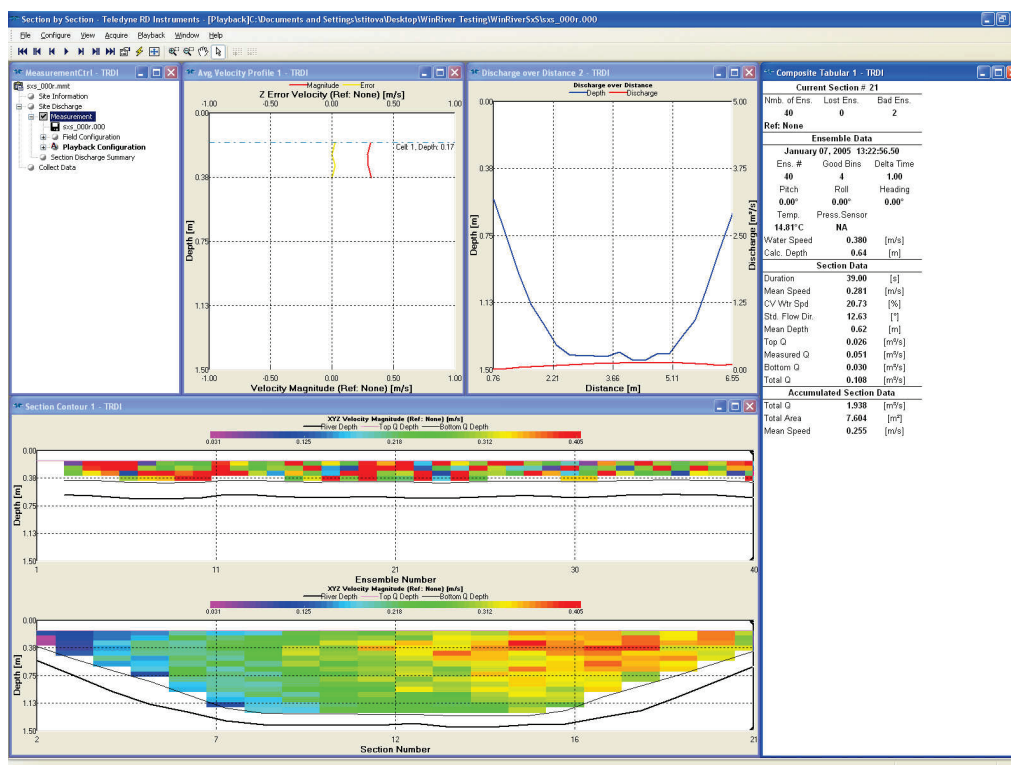


WinRiver II

Section-By-Section

User's Guide



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NOTES



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Section-By-Section User's Guide

1 Introduction

Section-By-Section is Teledyne RD Instrument's (TRDI) real-time discharge data collection program. This program creates a measurement file to operate the ADCP, checks each command, and verifies that the ADCP has received the commands.

Making accurate discharge measurements is less difficult than you probably believe now. You will soon see that you need to use only a few keystrokes on the computer to collect data in the field.

Please take the time to read this entire manual. It will be useful to have the ADCP and a computer available to follow along. You may also want to keep the other ADCP Technical manuals handy for reference when you want more detail.



NOTE. This guide covers *Section-By-Section* version 2.07.

1.1 How to Contact Teledyne RD Instruments

If you have technical issues or questions involving a specific application or deployment with your instrument, contact our Field Service group:

Teledyne RD Instruments

14020 Stowe Drive
Poway, California 92064

Phone +1 (858) 842-2600

FAX +1 (858) 842-2822

Sales – rdisales@teledyne.com

Field Service – rdifs@teledyne.com

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2A Les Nertieres
5 Avenue Hector Pintus
06610 La Gaude, France

Phone +33(0) 492-110-930

FAX +33(0) 492-110-931

Sales – rdie@teledyne.com

Field Service – rdiefs@teledyne.com

Client Services Administration – rdicsadmin@teledyne.com

Web: <http://www.rdstruments.com>

24 Hour Emergency Support +1 (858) 842-2700

1.2 Conventions Used in this Guide

Conventions used in the *Section-By-Section* User's Guide have been established to help you learn how to use these programs quickly and easily.

Windows menu items are printed in bold: **File** menu, **Save Measurement**. Items that need to be typed by the user or keys to press will be shown as <F1>. If a key combination were joined with a plus sign (<ALT+F>), you would press and hold the first key while you press the second key. Words printed in italics include program names (*BBTalk*) and file names (*Test_000_000.PD0*).

Code or sample files are printed using a fixed font. Here is an example:

```
WorkHorse Broadband ADCP Version X.XX  
RD INSTRUMENTS (c) 1996-2006  
ALL RIGHTS RESERVED  
>
```

You will find three other visual aids that help you: Notes, Cautions, and Recommended Settings.



NOTE. This paragraph format indicates additional information that may help you avoid problems or that should be considered in using the described features.



CAUTION. This paragraph format warns the reader of hazardous procedures (for example, activities that may cause loss of data or damage to the ADCP).



Recommended Setting. This paragraph format indicates additional information that may help you set command parameters.

1.3 Software Requirements

Section-By-Section requires the following:

- Windows Vista®, Windows XP Pro® or Windows 2000®
- Pentium III 600 MHz class PC (higher recommended)
- 64 megabytes of RAM (128 MB RAM recommended)
- 50 MB Free Disk Space plus space for data files (A large, fast hard disk is recommended)
- One Serial Port (two or more High Speed UART Serial Ports recommended)
- Minimum display resolution of 1024 x 768
- CD-ROM Drive
- Mouse or other pointing device

1.4 Software Installation

To install *Section-By-Section*, do the following.

- a. Insert the compact disc into your CD-ROM drive and then follow the browser instructions on your screen. If the browser does not appear, complete Steps “b” through “d.”
- b. Click the **Start** button, and then click **Run**.
- c. Type **<drive>:launch**. For example, if your CD-ROM drive is drive D, type **d:launch**.
- d. Follow the browser instructions on your screen.

1.5 Enabling the Section-By-Section Option

Prior to version 2.06, *Section-By-Section* required a Key Code to enable data collection. Now *Section-By-Section* requires a firmware feature to be installed in the ADCP to enable data collection. Contact TRDI Customer Service Administration (RDICSAdmin@teledyne.com) for the *Section-By-Section* firmware feature pricing and availability.

The firmware feature is dependent on the ADCP DSP serial number and can therefore be used on multiple computers running *Section-By-Section*. Note that *Section-By-Section* can read the old style key code, but will not be able to add new ADCPs unless they have the latest firmware and the firmware feature enabled.



NOTE. *Section-By-Section* data playback does not require a key code or firmware feature.

2 Overview of Section-By-Section

This section includes a brief overview of the inputs, displays, and outputs provided by *Section-By-Section*.

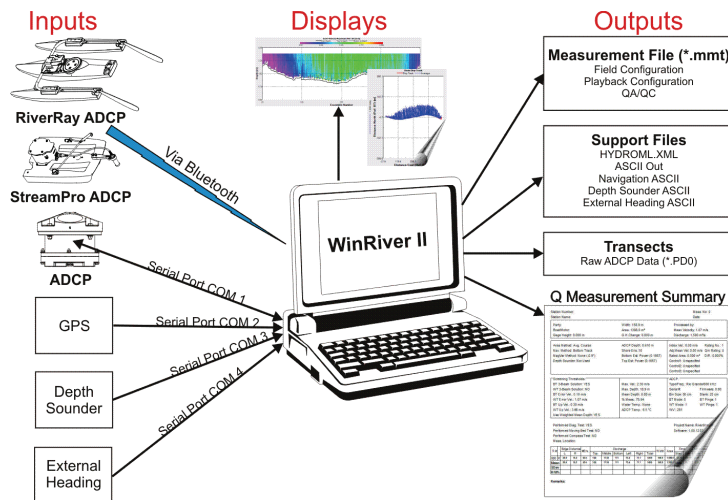


Figure 1. Overview of Section-By-Section

2.1 Comparison of WinRiver II and Section-By-Section

You will notice several similarities between *WinRiver II* and *Section-By-Section* and also several differences.

- Many of the displays are similar but some are unique to *Section-By-Section*. Note that when you are in *Section-By-Section*, the title bar at the top of the display is **Section-By-Section**. This makes it easy to see which program is running.
- In *WinRiver II*, the transect is done by crossing from bank to bank of the river. In *Section-By-Section*, transects are done by measuring at different user selected points on the river. This makes it possible to measure streams that are frozen over. This makes the **Discharge Summary** display different. *WinRiver II* shows the history of the entire transect; *Section-By-Section* shows the history of each section measured.
- Measurement files are different between *WinRiver II* and *Section-By-Section*. Use the proper program to play back data. A measurement data file created with *Section-By-Section* can not be used to play back data using *WinRiver II*.
- *WinRiver II* uses Bottom Tracking or GPS for velocity reference and *Section-By-Section* does not. This makes it possible to use

Section-By-Section software to measure discharge in streams with moving beds.

2.2 Inputs

In order to use *Section-By-Section*, your WorkHorse ADCP must meet the following criteria.

- WorkHorse Rio Grande ADCP or the WorkHorse ADCP **must** have the Bottom Track upgrade installed



NOTE. The Bottom Track and High Resolution Water Modes upgrades are available for WorkHorse Monitor and Sentinel ADCPs (included with the Rio Grande ADCP). *This guide assumes you will be using a WorkHorse Rio Grande ADCP.*

The Rio Grande, Monitor / Sentinel, and Mariner ADCPs can benefit from the upgrade to the High Rate Pinging (Water Mode 12) and Shallow Bottom Mode (Bottom Mode 7). Contact your local sales representative if you are interested in upgrading your system.

- *Section-By-Section* can also be used with a StreamPro ADCP if your computer has a Bluetooth connection.



NOTE. In StreamPro Firmware version 31.07 the Long Range mode has been extended to 6 meters. For compatibility with WinRiver II v2.06 and above, existing Long Range StreamPro users must have their firmware updated to version 31.07 (or higher).

ADCP Mounting Requirements

The ADCP must be mounted as follows.

- Over the side or through the hull mounted
- Downward facing
- Within 5 degrees of vertical (see note)

TRDI recommends using a fixed mount to achieve the best performance (see the WorkHorse Technical Manual for recommended sources of mounts).



NOTE. See the WorkHorse Commands and Output Data Format Guide for details on how the ADCP uses tilts in transformation (see the EX command).

2.3 Displays

All of the *Section-By-Section* displays may be saved to files or printed. For more information, see [Available Displays](#) and [Print a Graph or Display](#).

2.4 Outputs

The “heart” of *Section-By-Section* is the measurement file (*.mmt). A measurement file is created by running the **Measurement Wizard** (see [Using the Measurement Wizard](#)). The **Measurement Control** window helps keep track of the files used in the measurement and provides a quick way to access program controls by right-clicking on items in the list. For more information about the measurement control window, see [Using the Measurement Control Window](#).

Measurement File

- Configuration information – The **Field Configuration** for each measurement contains the settings used to collect the data. No changes can be made to this information once a measurement has started. A **Playback Configuration** is created when a measurement is reprocessed (see [Post-Processing of Discharge Data](#)).
- QA/QC information (ADCP tests, etc.)
- The **Measurement Control** window shows a list of all the measurement and support files.

Sections

- Raw ADCP data files (*.PDO) – These files contain all data sent from the ADCP during data collection. Refer to the WorkHorse Commands and Output Data Format guide for a complete description of the format of raw ADCP data files. For any specific deployment, raw data files contain the most information and are usually the largest.

Each time you start a section (using the **Acquire** menu or the shortcut key **F5**), *Section-By-Section* adds a section node to the **Measurement Control** window and opens a new *.PDO raw data file. The section number is incremented for each section. For more information on how to collect data, see [Acquiring Discharge Data](#).

The raw ADCP data files are saved to the same folder as the measurement file. Other TRDI software like *RDI Tools* may be used with the data files as needed. For more information about the format of the raw ADCP data, see the WorkHorse commands and Output Data Format guide.



CAUTION. TRDI's software *WinADCP* is not compatible for viewing and/or exporting *Section-By-Section* data, or data collected in Ship or Instrument coordinates. Using *WinADCP* to display and/or export *Section-By-Section* data results in an incorrect naming structure. Data is called Earth coordinates when it is actually Ship coordinates.

Q Measurement Summary

- The **Q Measurement Summary** is a printed report of the discharge. See [Using the Q Measurement Summary](#) for more information.

3 Communications Setup

When *Section-By-Section* is first started, you can set up the communications with the ADCP. Once set up, *Section-By-Section* will remember the settings and use them each time the program is started.



NOTE. Using the **Measurement Wizard** (see [Using the Measurement Wizard](#)) will also prompt you to set up communications if they are not already configured.

- Connect and power up the ADCP as shown in the appropriate ADCP User's Guide.
- Start *Section-By-Section*. To open the **Peripheral Configuration Dialog** box, on the **Configure** menu, select **Peripherals**.

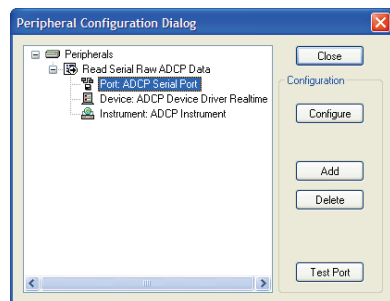
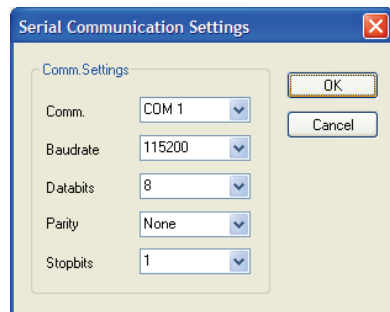


Figure 2. Peripheral Configuration Dialog

- Click the + box next to **Read Serial Raw ADCP Data** to expand the list and then select **Port: ADCP Serial Port**. Press the **Configure** button.



NOTE. The default baud rate for WorkHorse ADCPs is 9600. RiverRay and StreamPro ADCPs use 115200 baud. Use the **Advanced ADCP Communications Configuration** screen (see Figure 3) to set the ADCP's

baud rate as high a rate as possible and configure *Section-By-Section* to match the ADCP's baud rate.

- d. Select the **Communication Port, Baudrate, Databits, Parity, and Stopbits**. If you are unsure of the setting, click **Cancel** and right-click on **Instrument: ADCP Instrument** and select **Auto Detect**. Click **OK** to continue. Click **Test Port** to have *Section-By-Section* connect to the device and confirm the communication setting.
- e. Once set up, select **Close** to exit the **Peripheral Configuration Dialog** box.

3.1 StreamPro & RiverRay Communication Setup

Use these next steps to setup a Bluetooth connection to a RiverRay or StreamPro ADCP.

- a. Turn on the power to the ADCP.
- b. Configure your computer's Bluetooth device. It should locate the RiverRay or StreamPro ADCP (**RDI RRay 00xxx** or **RDI SPro 00xxx** where xxx is the serial number).



NOTE. Some Bluetooth configurations require a pin code for StreamPro and RiverRay ADCPs. The pin code is 0.

- c. Double-click on the **RDI RRay** or **RDI SP** icon and make sure that it shows **Connected**.
- d. Make note of the **Com Port** number it is using.
- e. On the *WinRiver II* **Configure** menu, select **Peripherals**.
- f. Select **Port: ADCP Serial Port** and then click the **Configure** button.
- g. Enter the **Comm. Port** number as noted in the Bluetooth screen.

3.2 Advanced ADCP Communications Configuration

Use the following steps to communicate with the ADCP or verify command settings.

- a. On the **Configure** menu, select **Peripherals**.
- b. Click the + box next to **Read Serial Raw ADCP Data** to expand the list and then select **Instrument: ADCP Instrument**.
- c. Press the **Configure** button.

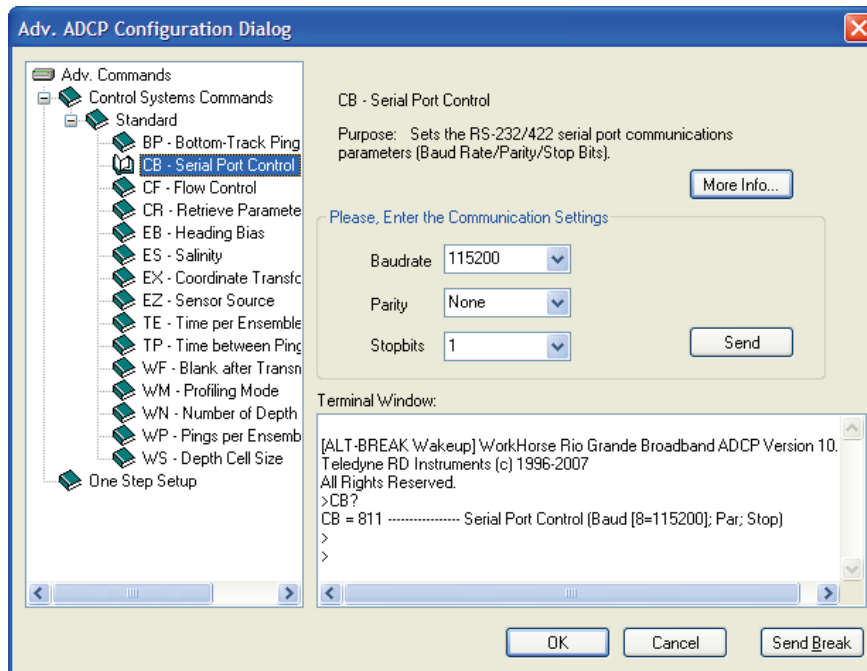
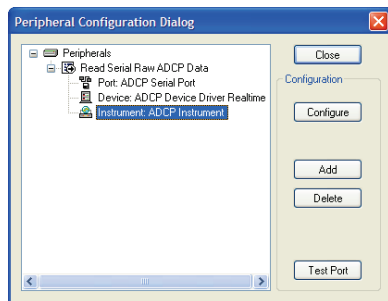


Figure 3. Advanced ADCP Configuration Dialog

- d. Use the **Advanced ADCP Configuration Dialog** screen to send or verify command setting as needed. Click the **Send** button to send the command to the ADCP and view the results of the command in the **Terminal Window**.
- e. Use the **Send Break** button to wake up the ADCP.
- f. If the command you want to send the ADCP is not on the list, type it next to the ">" prompt in the **Terminal Window** and press return.



CAUTION. Do not use this screen to deploy or configure the ADCP. The commands will be overwritten when **Acquire, Start Pinging (F4)** is pressed. Only the commands generated through the wizard and sent via **Acquire, Start Pinging (F4)** can be used to deploy the ADCP.



NOTE. WorkHorse Monitor/Sentinel ADCPs will go to sleep if a command is not sent within five minutes; Use the **Send Break** button to wake the

ADCP before sending a command.

WorkHorse Rio Grande ADCPs do not sleep. For more information, see the CL command in the WorkHorse Commands and Output Data Format guide.

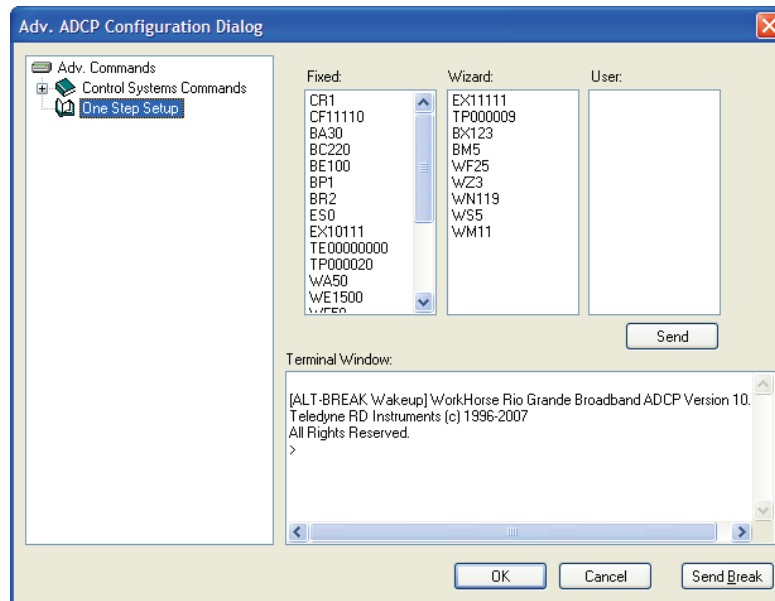


Figure 4. One Step Setup

One Step Setup

Use the **One Step Setup** to check the deployment commands sent to the ADCP after the **Measurement Wizard** has been completed (see [Using the Measurement Wizard](#)).

- Click **One Step Setup**. This will show you the commands that have been generated by the Wizard and are ready to be sent to the ADCP when you start collecting data.
- Click the **Send** button. If any command generates an error message, stop and correct the problem before deploying.



CAUTION. Do not use this screen to deploy or configure the ADCP.

The commands will be overwritten when **Acquire, Start Pinging (F4)** is pressed. Only the commands generated through the wizard and sent via **Acquire, Start Pinging (F4)** can be used to deploy the ADCP.



NOTE. The **One Step Setup** is a good way to check if any command generates an error message (for example, if the commands depend on the High Resolution Water Profiling upgrade but you are not sure the ADCP has this feature).

If a command generates an error, a message box will appear.

The **One Step Setup** does not check the commands for “reasonableness”. If the wizard has not been run, the **Fixed** and **Wizard** command boxes will be blank.

Commands can be entered in the **User** box or at the “>” prompt.

4 Customizing Section-By-Section

Section-By-Section can be customized to look and act, as you prefer.

4.1 Workspace

A Workspace is a collection of windows arranged and sized, as you prefer. To create a Workspace file, open all the windows you want to see during data collection or post-processing. Open and arrange the views you are interested in. When you have the displays set up the way you prefer, on the **File** menu, click **Save Workspace As**. To use a Workspace file, on the **File** menu, click **Load Workspace**. How workspaces are loaded and saved when starting/closing *Section-By-Section* depends on the **User Options** (see [General Configuration](#)).

4.2 User Options

On the **File** menu, click **Properties**. The **Properties** dialog sets how *Section-By-Section* behaves upon entering the Acquire mode and how workspace files are loaded or saved.

4.2.1 Acquire Mode Properties

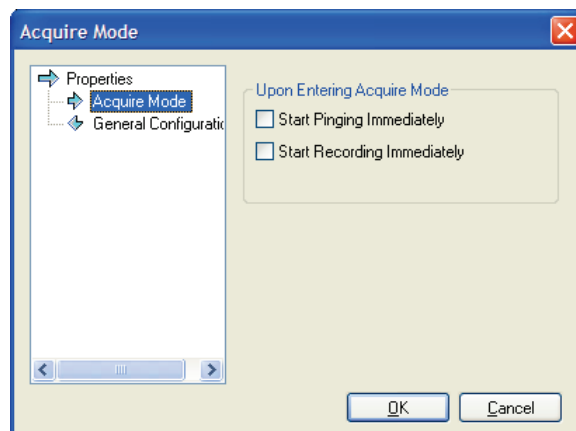


Figure 5. User Options – Acquire Mode
Upon Entering Acquire Mode

- **Start Pinging Immediately** – Select this option if you want the ADCP to begin pinging as soon as the Acquire mode is started.
- **Start Recording Immediately** – Select this option if you want the ADCP to begin recording as soon as the Acquire mode is started.

4.2.2 General Configuration

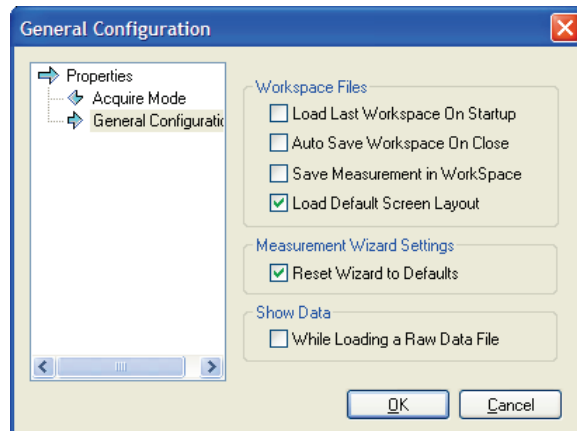


Figure 6. User Options – General Configuration

Workspace Files

- **Load Last Workspace On Startup** – Select this option if you want the same graphs and displays opened as soon as *Section-By-Section* is started.
- **Auto Save Workspace On Close** – Select this option if you want to automatically save any changes to the workspace whenever *Section-By-Section* is exited.
- **Save Measurement in Workspace** – Select this option to include the measurement file with the workspace.
- **Load Default Screen Layout** – Check this box to use the default workspace.

Measurement Wizard Settings

Check the **Reset Wizard to Defaults** box to have the measurement wizard use the default settings based on the frequency of the user's ADCP.

Show Data

Check the **While Loading a Raw Data File** box to begin displaying data while the file is loading. For smaller data files, this may not be noticeable.

4.3 Display Options

The following options are available for all graphs.



NOTE. All graph scales and font sizes are user-selectable in real-time by right-clicking on the graph and selecting **Properties**.

4.3.1 Changing Units

To change the units for all displays, on the **Configure** menu, select **Units**. You can change units to **All English**, **All SI** or use the **Advanced** menu and select each unit for **Velocity**, **Range/Depth**, and **Temperature**.



NOTE. When using a StreamPro ADCP on small streams/channels, change the Velocity units to mm/s or cm/sec. This will change the discharge to mm³/s or cm³/s.

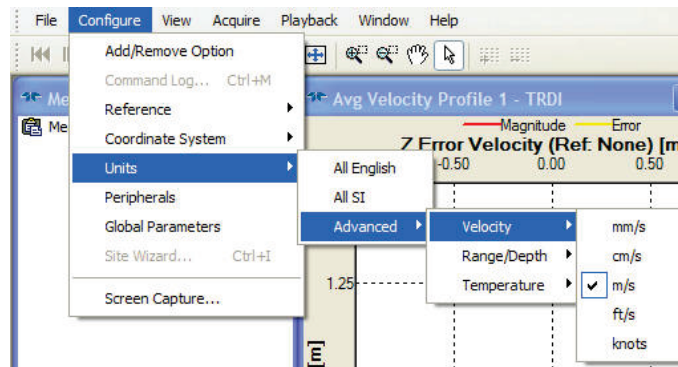


Figure 7. Changing the Units

4.3.2 Changing the Reference

To change the reference, on the **Configure** menu, select **Reference**. You can select one of the following references.

- **Bottom Track** – ADCP Bottom-Track velocity (standard configuration)
- **None** – No reference is used.

4.3.3 Coordinate System

To change the coordinate system for the displays, on the **Configure** menu, select **Coordinate System**. The default coordinate system is **Earth** for Rio Grande ADCPs and **XYZ** for StreamPro ADCPs.

4.3.4 Global Parameters

To change the global parameters of the displays, select the **Configure** menu, and click **Global Parameters**. This menu allows changes to the number of sections on the X and Y axis, the Grid Lines (visible, style, and color), and Background Color. To return to the factory defaults and the default window layout, check the **Reset to Factory Defaults** box and select **Apply**. Click **OK** to exit the screen.

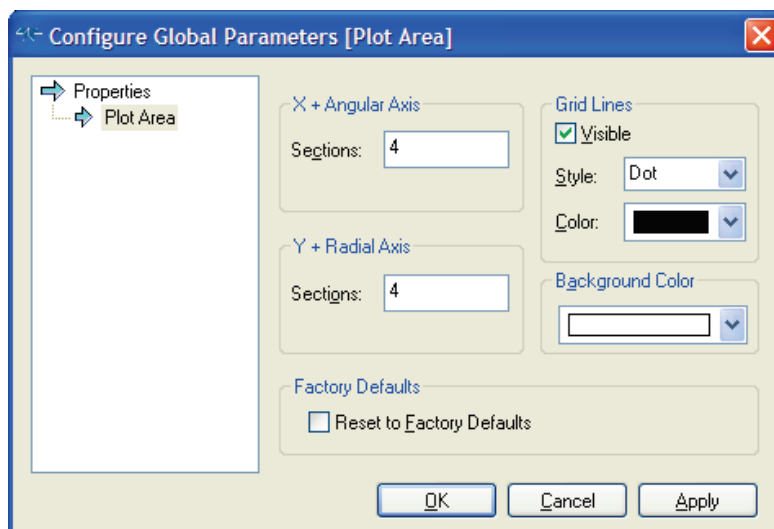


Figure 8. Global Parameters for Graphs

4.3.5 Zoom Functions

- **Zoom to Extents** – click the blue title bar at the top of the window to select the graph and then click the **Zoom to Extents** button on the toolbar (). The data will be zoomed to the full graph width.
- **Zoom In** – Click the Zoom In () button on the toolbar. The cursor will change to a “+” magnifying glass. Hold down the left mouse button and drag over an area on the window to zoom in. The chosen region will be zoomed to the full graph width.
- **Zoom Out** – Click the Zoom Out () button on the toolbar. The cursor will change to a “-” magnifying glass. Hold down the left mouse button and drag over an area on the window to zoom out.
- **Pan** – Use the **Pan** button on the toolbar () to move the graph data as needed.
- **Arrow** – Use the Arrow button on the toolbar to identify the data under the cursor point.

5 Using the Measurement Control Window

The **Measurement Control Window** gives the user a quick and easy way to manage the files used in a measurement. It uses a tree structure; click the + box to expand the list or – to collapse. Right-clicking on a name will bring up menus to quickly access different functions.

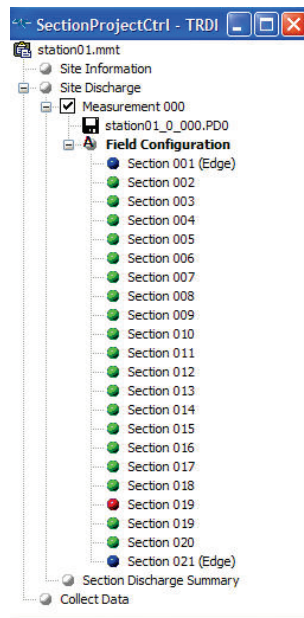


Figure 9. Measurement Control Window

5.1 Measurement File Name

Right-click on the measurement file name to display the following menu. If a measurement file is not already open, only the **New Measurement**, **Create Measurement from Data Files**, and **Open Measurement** menu items are available.

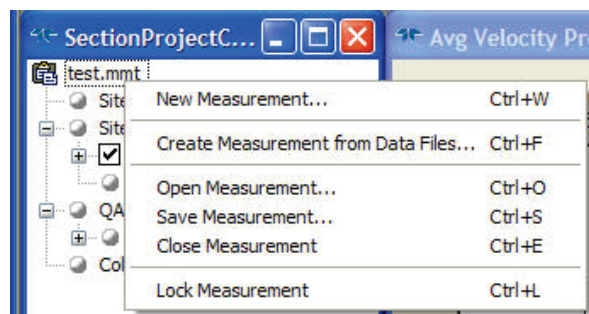


Figure 10. Measurement Control - Measurement Menu

5.1.1 New Measurement

Select **New Measurement** to start the Measurement Wizard (see [Using the Measurement Wizard](#)). If a measurement is already open, you will be prompted to save the current measurement.

5.1.2 Create Measurement from Data Files

Use **Create Measurement from Data Files** and select one data file to create a measurement. The file must have been collected with the *Stream-Pro* software with the *Section-By-Section* feature enabled.

5.1.3 Save Measurement

Use the Measurement Wizard to set up *Section-By-Section* (see [Using the Measurement Wizard](#)). Once the Measurement Wizard is completed, on the **File** menu, select **Save Measurement**. Enter the **File name** and select **Save**. *Section-By-Section* will automatically add the file extension *.mmt.



NOTE. *Section-By-Section* automatically saves the measurement file each time you start/stop a section.

5.1.4 Open Measurement

To open a measurement file, on the **File** menu, select **Open Measurement**. On the **Select Measurement File** dialog, select the measurement file and click **Open**.

5.1.5 Close Measurement

To close a measurement file, on the **File** menu, select **Close Measurement**. On the **Close Measurement Confirmation** dialog, select **Yes** to close the measurement file.

5.1.6 Lock Measurement

Use **Lock Measurement** to write protect the measurement file. If a measurement file is locked, any changes made to the playback configuration nodes will not be saved unless the measurement file is first “unlocked”. This prevents accidental saving of changes made to configuration nodes while “experimenting” with different settings. To lock a measurement file, right-click on the **measurement file name** and select **Lock Measurement**. To unlock a measurement file, right-click on the **measurement file name** and select **Unlock Measurement**. The icon next to the measurement file name will change from a lock to a clipboard to let you see if it is locked or not.

5.2 Site Information

Right-click on **Site Information** and select **Site Wizard**; this opens the wizard and allows changes to the **Site Information** (see [Site Information](#)) page of the wizard. If changes are made, save the measurement file (see [Save Measurement](#)).

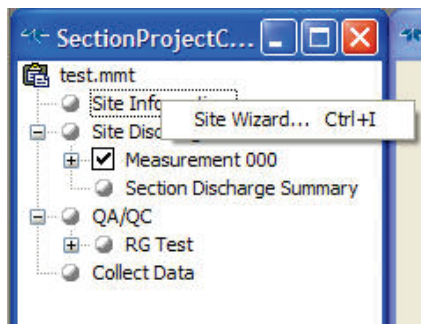


Figure 11. Measurement Control - Site Information Menu

5.3 Measurement

Right-clicking on **Measurement** will display the following menu.

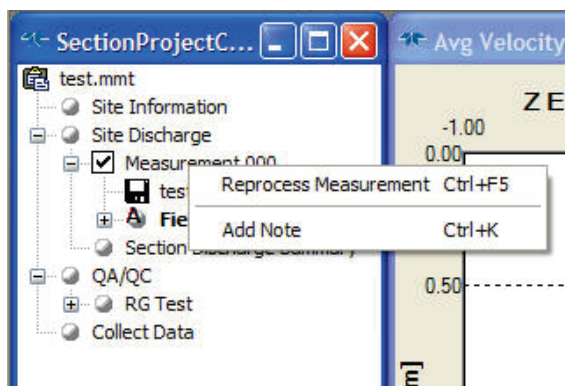


Figure 12. Measurement Control – Transect

5.3.1 Reprocess Measurement

Use this function to reprocess the measurement file.

5.3.2 Add Note

Use this function to add a note to the measurement file for the selected transect. For example, you may want to record a note about the instrument setup or factors such as wind conditions, the passage of other vessels, and any other noteworthy events that occur during your transect of the channel.

5.3.3 Data File

Right-clicking on a data file and selecting **Properties** will display the **Raw Data File Properties** box.

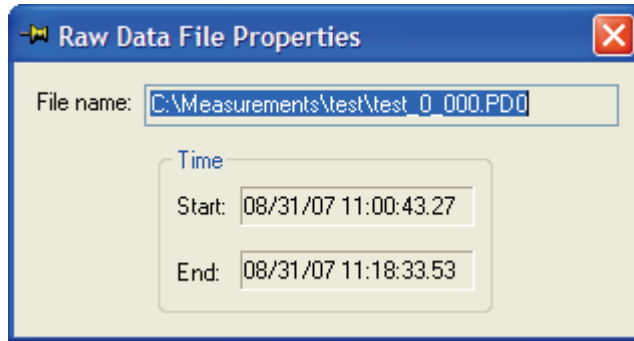


Figure 13. Measurement Control - Raw Data File Properties

5.4 Field and Playback Configuration Nodes

Once you reprocess a transect (see [Reprocess Measurement](#)) or duplicate the **Field Configuration** node, *Section-By-Section* creates a **Playback Configuration** node. The **Playback Configuration** is a copy of the **Field Configuration** information used to collect the data. Any editing changes made to the **Playback Configuration** are saved to the measurement file. **Playback Configuration** nodes can be **Duplicated**, **Deleted**, and **Renamed** as needed.

5.4.1 Configuration Node Menu Options

Right-clicking on a **Field Configuration** or **Playback Configuration** node will display the following menu.

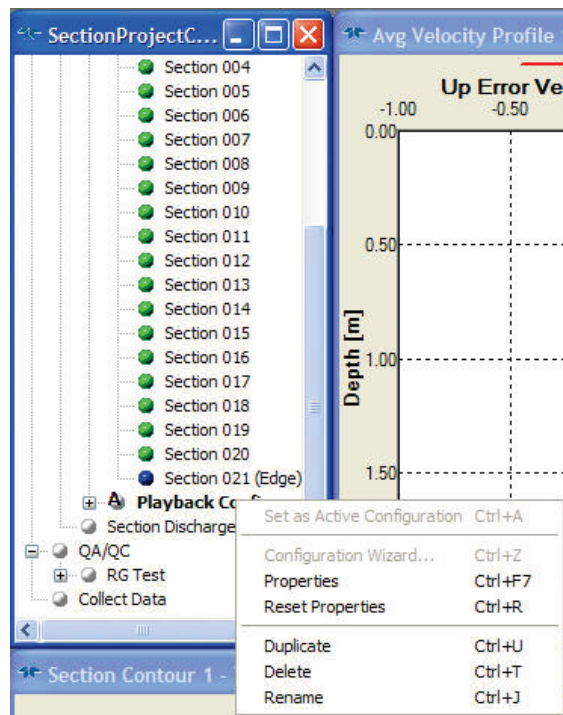


Figure 14. Managing Configuration Nodes

5.4.2 Set as Active Configuration

The active configuration node is in bold. Right-click the on the configuration node name and select **Set as Active Configuration** to change between configuration nodes. Only one configuration node may be active at a time.

5.4.3 Configuration Wizard

This option is only available in the Acquire mode and only if no measurements have been collected.

- a. On the **Acquire** menu, select **Stop Pinging**.
- b. Right-click on the **Field Configuration** node and select **Configuration Wizard**; This will restart the Measurement Wizard (see [Configuration Dialog](#)).
- c. Make your adjustments.
- d. To restart, click the **Acquire** menu and select **Start Pinging**.



NOTE. Once a measurement has started, the field configuration node can no longer be changed.

5.4.4 Properties

Use this function to view the configuration settings (see [Configuration Node Properties](#)). Changes to the settings are only allowed on **Playback Configuration** nodes or if data collection for the transect has not started. You can view what settings were used for the **Field Configuration**.

5.4.5 Reset Properties

This resets the playback configuration node back to the same parameters as the field configuration node.

5.4.6 Duplicate

Use this function to duplicate the configuration node. If the Field Configuration node is duplicated, it will create a new Playback Configuration node.

5.4.7 Delete

Use this function to delete the configuration node. The Field Configuration node can not be deleted.

5.4.8 Rename

Use this function to rename the configuration node. The Field Configuration node can not be renamed.

5.5 Show Summary

Right-click on the **Section Discharge Summary** field and select **Show Summary** to display the **Discharge Summary** (see [Section Discharge Summary](#)).

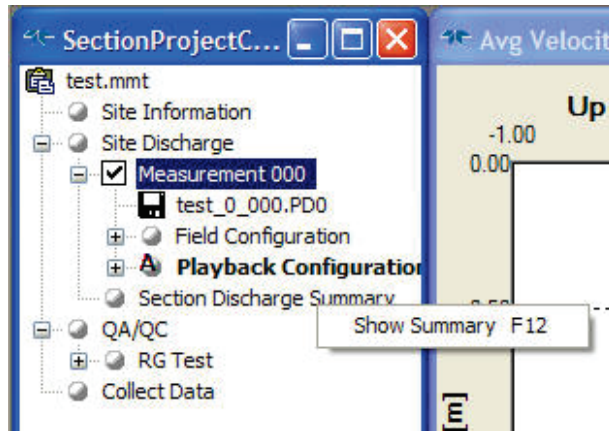


Figure 15. Measurement Control - Discharge Summary

6 Available Displays

Use the **View** menu to select what graphs and tabular data to display. *Section-By-Section* can use the following screens.

- Measurement Control Window (see [Using the Measurement Control Window](#))
- Contour Graphs
- Profile Graphs
- Tabular Displays

6.1 Contour Graphs

The **View, Graphs, Contour, Velocity** menu lets you select the type of velocity contour graph to display.



NOTE. To change the graph scale, right-click on the graph and select **Properties**.

To change what data is displayed, right-click on the graph and select **Data Selection**.

The available velocity contour graphs are the four earth-referenced velocity components (East, North, Up, Error, Magnitude, and Direction).

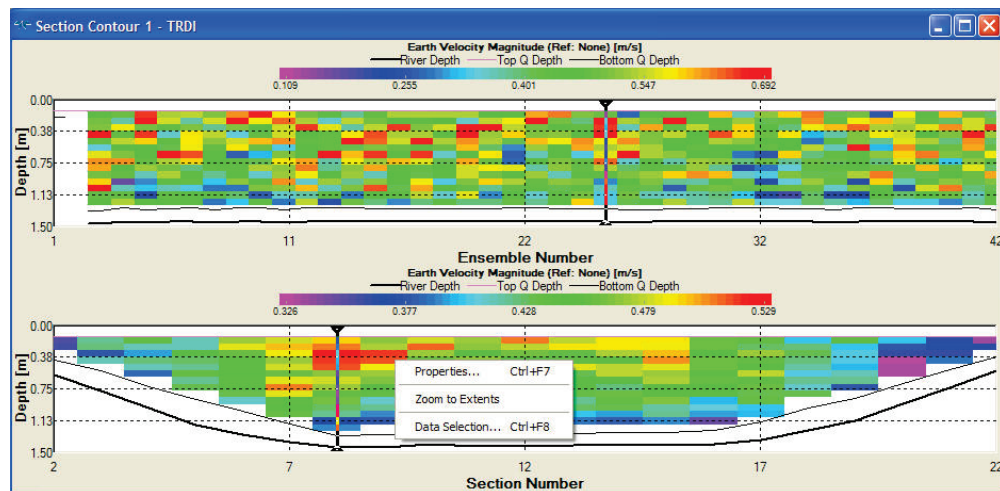
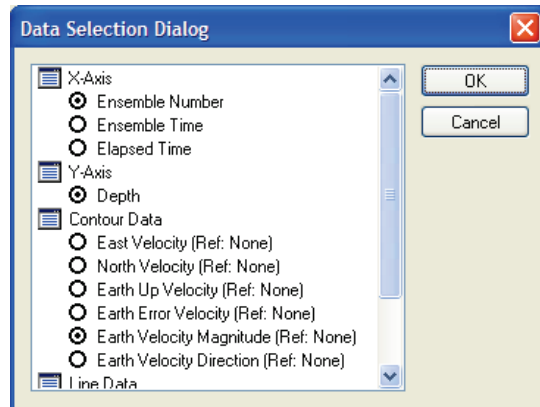


Figure 16. Velocity Contour Graph

The following are the velocity contour graph options.



- **East Velocity** – East is positive (west is negative).
- **North Velocity** – North is positive (south is negative).
- **Up Velocity** – Up is positive (down is negative).
- **Error Velocity** – This velocity component is a measure of “data reasonableness.” There is no velocity reference for this value. The error velocity calculation depends on transducer orientation and beam pattern. See the ADCP Technical Manual for details of the velocity processing algorithm.
- **Velocity Magnitude/Direction** – The velocity magnitude and direction are clockwise with respect to North.

Click **View, Graphs, Contour, Intensity** to select the echo intensity contour graph.

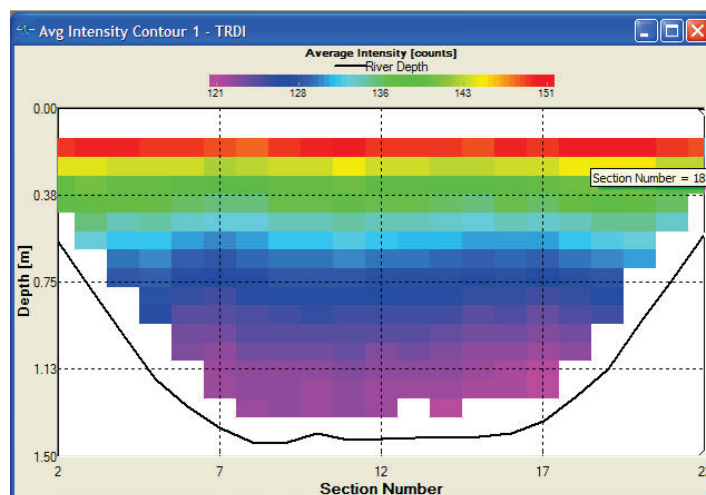
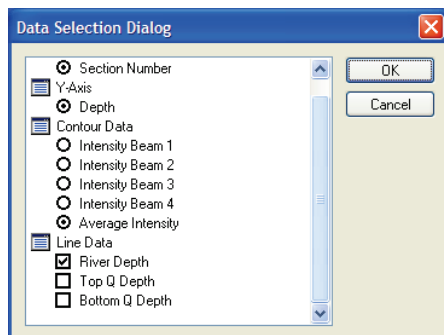


Figure 17. Intensity Contour Graph

To change what data is displayed, right-click on the graph and select **Data Selection**. The five intensity graphs are: **Beam 1**, **Beam 2**, **Beam 3**, **Beam 4**, and **Average**.



6.2 Profile Graphs

A profile graph is a line graph of a selected parameter versus depth. The **View, Graphs, Profile** menu lets you select the type of profile graph to display.



NOTE. To change the graph scale, right-click on the graph and select **Properties**.

To change what data is displayed, right-click on the graph and select **Data Selection**.

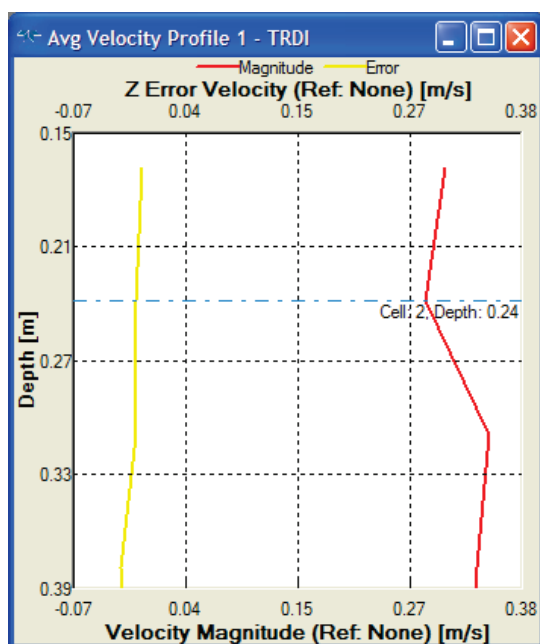


Figure 18. Velocity Profile Graph

The following profile graphs are available.

- **Intensity Profile** – *Section-By-Section* displays the intensity profiles for all four ADCP beams. The echo intensity units are in counts. Use the Intensity minimum/maximum scaling values to define the horizontal axes of the graph.
- **Velocity Profile** – The velocity profiles show the earth-coordinate section average velocity components (East, North, Up, and Error). The velocity units can be in either m/s (SI) or ft/s (English). Scaling for the top horizontal axis on the graph (Up Error Velocity) uses the minimum/maximum scale values of the Up Velocity or Error Velocity. Similarly, the bottom horizontal axis (East North Velocity) uses either the East Velocity or North Velocity scale values.
- **Backscatter Profile** – *Section-By-Section* displays the backscatter profiles for all four ADCP beams. Backscatter data is range and absorption normalized and the units are in decibels (dB). Data are obtained from the receiver's Received Signal Strength Indicator (RSSI) circuit. The default of **Calculate for each Ping** is used (similar to *WinRiver II*), but in *Section-By-Section* it can not be changed at this time.
- **Correlation Profile** – *Section-By-Section* displays the correlation profiles for all four ADCP beams. Correlation is a measure of data quality, and its output is scaled in units such that the expected correlation (given high signal/noise ratio, S/N) is 128 for Mode 1.
- **Section Velocity Magnitude** – Displays the velocity magnitude of all section ensembles, section mean velocity, and its model.

6.3 XY Graphs

Discharge over Distance – This XY graph displays the horizontal measured discharge and corresponding power fit versus depth.

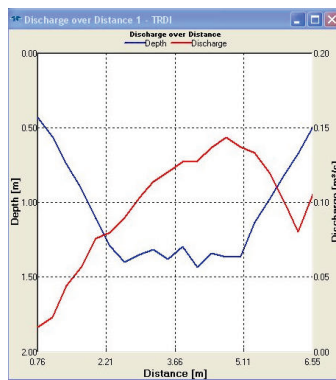


Figure 19. Discharge over Distance Graph

6.4 Tabular Displays

The Tabular displays show the data for the last ensemble in tabular form. *Section-By-Section* displays as **Bad** the data flagged bad by the ADCP.

Table 1: Available Tabular Displays

Display	Description
Ensemble Header	The Ensemble Header display shows information about the ADCP setup.
Composite	The Composite display lists information about the currently displayed ADCP data ensemble, and discharge information.
Bottom Track	The Bottom Track display lists information about the beam depths and bottom track velocities.
Velocity	Displays the earth-referenced velocity components (East, North, Up, Error) percent good pings for the bin, and discharge. <i>Section-By-Section</i> displays as bad the data flagged bad by the ADCP.
Earth Velocity Magnitude & Direction	Displays the earth-referenced velocity components (Magnitude, Direction, Up, Error) percent good pings for the bin, and discharge.
Intensity	Displays the echo intensities for all four beams.
Backscatter	Displays the echo intensities for all four beams.
Correlation	Displays the correlation magnitudes for all four beams.



NOTE. During data collection, the text color will change to blue when the file is no longer being updated.

6.5 Section Discharge Summary

Select **View, Section Discharge Summary** to display the Section Discharge Summary. This screen displays detailed information and statistics about each section.

Measurement	Q [m³/s]	% Total Q	Accum Q [m³/s]	% Diff. Rating	% Rated Q	Avg. Velocity [m/s]	Coeff. Var [%]	Width [m]	Distance [m]	Depth [m]	Area [m²]	Flow Angle Coeff.
Measurement ...	1.938			340.34		0.255		7.438			7.600	
Section 001	0.000	0.00%	0.000		0.00	0.000	0.00	0.366	0.030	0.000	0.000	1.000
Section 002	0.017	0.86%	0.017		3.77	0.059	60.13	0.518	0.762	0.546	0.283	1.000
Section 003	0.023	1.20%	0.040		5.26	0.099	43.69	0.305	1.067	0.764	0.233	1.000
Section 004	0.044	2.30%	0.084		10.11	0.153	29.57	0.304	1.372	0.957	0.291	1.000
Section 005	0.057	2.94%	0.141		12.95	0.161	25.36	0.304	1.676	1.159	0.353	1.000
Section 006	0.076	3.91%	0.217		17.24	0.196	22.89	0.305	1.981	1.271	0.388	1.000
Section 007	0.090	4.11%	0.297		18.23	0.193	20.57	0.305	2.286	1.360	0.415	1.000
Section 008	0.090	4.65%	0.387		20.46	0.209	18.21	0.305	2.591	1.415	0.431	1.000
Section 009	0.103	5.34%	0.491		23.51	0.239	11.96	0.304	2.896	1.423	0.433	1.000
Section 010	0.114	5.89%	0.605		25.92	0.263	17.19	0.304	3.200	1.422	0.433	1.000
Section 011	0.120	6.21%	0.725		27.36	0.277	13.99	0.305	3.505	1.425	0.434	1.000
Section 012	0.127	6.57%	0.852		29.91	0.291	10.76	0.305	3.810	1.433	0.437	1.000
Section 013	0.128	6.59%	0.980		29.03	0.290	11.66	0.305	4.115	1.444	0.440	1.000
Section 014	0.137	7.06%	1.117		31.08	0.310	10.12	0.304	4.420	1.447	0.441	1.000
Section 015	0.144	7.42%	1.261		32.66	0.327	12.24	0.304	4.724	1.442	0.439	1.000
Section 016	0.137	7.08%	1.398		31.19	0.320	9.86	0.305	5.029	1.406	0.429	1.000
Section 017	0.133	6.88%	1.531		30.31	0.333	11.48	0.305	5.334	1.311	0.400	1.000
Section 018	0.120	6.18%	1.651		27.22	0.321	10.04	0.305	5.639	1.224	0.373	1.000
Section 019	0.101	5.21%	1.752		22.94	0.319	15.30	0.304	5.944	1.038	0.316	1.000
Section 020	0.081	4.17%	1.833		18.34	0.321	15.86	0.304	6.248	0.826	0.251	1.000
Section 021	0.006	0.31%	1.839		23.65	0.277	20.56	0.610	6.553	0.630	0.376	1.000
Section 022	0.000	0.00%	1.938		0.00	0.000	0.00	0.457	7.468	0.000	0.000	1.000

Figure 20. Discharge Summary Screen

6.6 Acquire Control Window

Select **View**, **Acquire Data** to display the Section Acquire Control window. This screen gives you a quick view to see if the ADCP is recording or pinging.

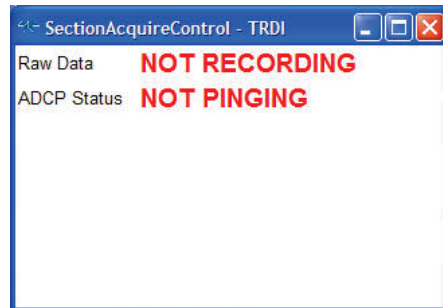


Figure 21. Section Acquire Control Window

6.7 Adding Customized Graphs to the Menu

Please note that not all available graphs are shown on the menu. For example, the profile graph **Velocity Magnitude and Direction** is not predefined graph. On the **View** menu, select **Graphs, Profile, Velocity**. Right-click on the velocity profile graph and select **Data Selection**. Select **Earth Velocity Magnitude** and **Earth Velocity Direction** and then click **OK**.

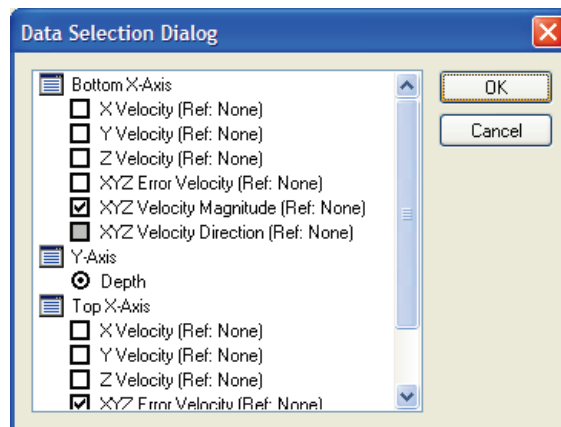


Figure 22. Data Selection Dialog

Once you have selected a graph and selected the data you want to see, right-click on the graph and select **Add to Menu**. Name the graph and click **OK**. The graph will now be available on the menu for future use with a click of the mouse button. If you want to always see the graph when *Section-By-Section* starts, make sure to save the workspace by clicking **File, Save Workspace As**.

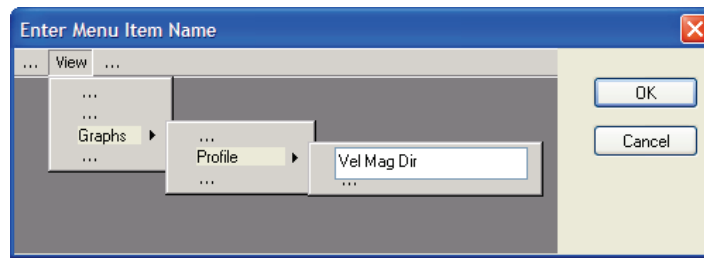


Figure 23. Adding a Graph to the Menu



NOTE. There is no method to remove a graph once added to the menu except by re-installing *Section-By-Section*.

7 Using the Measurement Wizard

Section-By-Section is set up to collect data using the **Measurement Wizard**. The **Measurement Wizard** gives the user the ability to create a Field Configuration node and enter the information needed for data collection and correct data display during data collection. Once set up, the measurement file can be saved and then later retrieved.

- On the **File** menu, click **New Measurement**.
- After each page information has been entered, click **Next**.
- After reviewing the **Summary** page, click the **Finish** button to complete the wizard.

7.1 Site Information

Figure 24. Site Information

On the **Site Information** screen, enter the **Station Data**, **Agency Data** and **Field Party Data** information. Each of these fields will be included on the Q Measurement Summary file (see [Using the Q Measurement Summary](#)).

Enter the Station Data **Stage (m)**, **Rated Q (m³/s)** and **Tail Water Level (m)** information.



NOTE. If a **Station Number** is entered, it will be used in the **Filename Prefix** box on the **Output Filename Options** page of the wizard (see [Output Filename Options](#)).

Enter a **Measurement Number** (alphanumeric). This can be added to the file name on the **Output Filename Options** page of the wizard.

7.2 Configuration Dialog

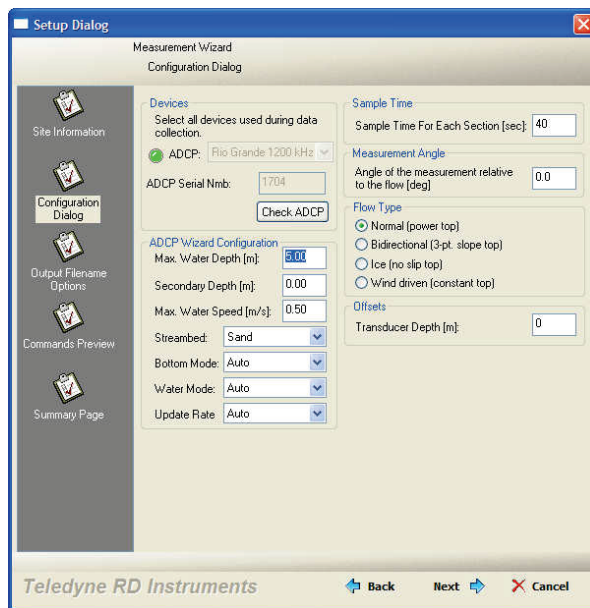
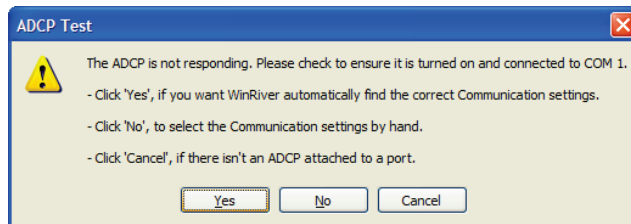


Figure 25. Configuration Dialog

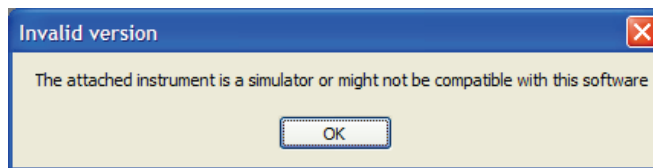
7.2.1 Devices

Section-By-Section will attempt to connect to the ADCP when this page opens or the **Check ADCP** button is clicked. If the ADCP is not available or communications with the ADCP have not been set up, you will see the following dialog box.



Click **Cancel** to continue with the wizard, click **Yes** to have *Section-By-Section* automatically detect the ADCP, or click **No** to set up the communications yourself.

If you see the following message box, this means *Section-By-Section* is not sure what type ADCP you are using. Click **OK** to continue.



If you are using a Rio Grande ADCP, *Section-By-Section* will automatically detect the ADCP and enter the **ADCP Serial Number**, otherwise, enter the serial number. The ADCP serial number will be added to the Q Measurement Summary (see [Using the Q Measurement Summary](#)).

7.2.2 ADCP Wizard Configuration

Enter your choices for the **ADCP Wizard Configuration** section. Based on the entered information, the wizard will enter commands on the [Commands Page](#). *Section-By-Section* will give warning messages if the settings are not recommended. Click the **Close** button on the message box or continue making choices.

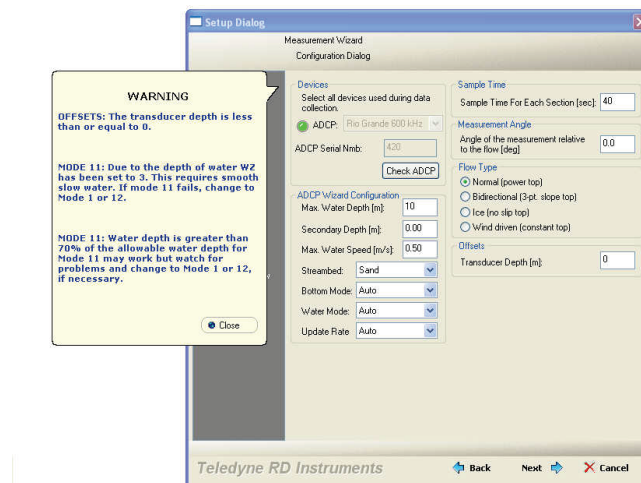


Figure 26. Warning Message

Max. Water Depth – Enter the expected maximum depth of the stream.

Secondary Depth – Enter a secondary depth (optional). For example, enter the minimum depth at the edge.

Max. Water Speed – Enter the expected maximum speed of the stream.

Max. Boat Speed – Enter the expected maximum speed of the boat.

Streambed – Enter the expected streambed material.

Bottom Mode – Enter the bottom mode (**Mode 5**, **Mode 7**, or **Auto**).



Recommended Setting. Use the **Auto** mode.

Water Mode – Enter the water mode (**Mode 1**, **Mode 5**, **Mode 8**, **Mode 11**, **Mode 12 Auto**, **Mode 12 Custom**, or **Auto**).



Recommended Setting. Use the **Auto** mode.

If you select **Mode 12 Custom**, the following screen appears.

Figure 27. Water Mode 12 Options

Bin Size (meters) – Set the desired bin size. This will add the WK command to the Wizard section of the command page.

Update Rate (seconds) – Increasing the **Update Rate** will increase the number of subpings.

Time Between Subpings (100th seconds) – This set the time between subpings in hundredths of a second and sets the second parameter of the WO command. In the above example, the WO command would be set to WO13,5.

Click the **Compute** button to see the **Optimized Configuration** consequences. Click **Done** to return to the **Configuration Dialog** screen.



NOTE. Your ADCP must have the High Rate Ping feature enabled in order to use Mode 12.

7.2.3 Sample Time

The default **Sample Time** is 40 seconds. You may increase this if desired. In slow, shallow flow increasing this value will reduce the uncertainty of the measured velocity.



Recommended Setting. Leave the Sample Time at 40 seconds.

7.2.4 Measurement Angle

If the measurement line is not perpendicular to the flow, then enter the angle of deviation from perpendicular in the **Measurement Angle** box. This will reduce the discharge to compensate for the fact that each panel is wider than would be the case if the measurement line were perpendicular.

7.2.5 Flow Type

Normal is selected by default for the **Flow Type**. *Section-By-Section* will use power extrapolation at the surface and power extrapolation to the bottom.

If the velocity profile is expected to be non-classical, wind driven at the surface for example, then selecting the **Bidirectional, wind driven** box will use 3-Point Slope extrapolation at the surface and No Slip extrapolation to the bottom.

If you are measuring below ice cover, check the **Ice** box and enter the **Transducer Depth** from bottom of the ice cover to the center of the ADCP transducer faces. *Section-By-Section* will use No Slip extrapolation at the surface and No Slip extrapolation to the bottom.



NOTE. For a full description of the methods used to estimate discharge in the unmeasured top/bottom parts of the velocity profile, see the WinRiver II User's Guide (Appendix B - Discharge Measurement Basics).

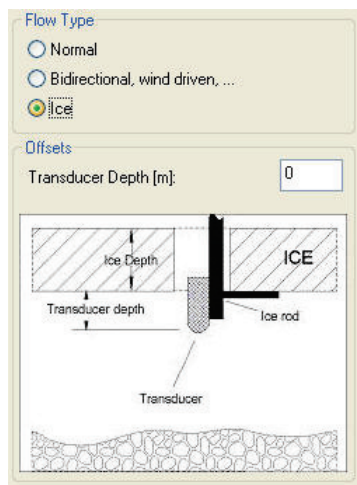


Figure 28. Ice

7.2.6 Offsets

Enter your choices for the **Offsets** section.

Transducer Depth – Enter the depth from the water surface to the ADCP transducer faces.

7.3 Output Filename Options

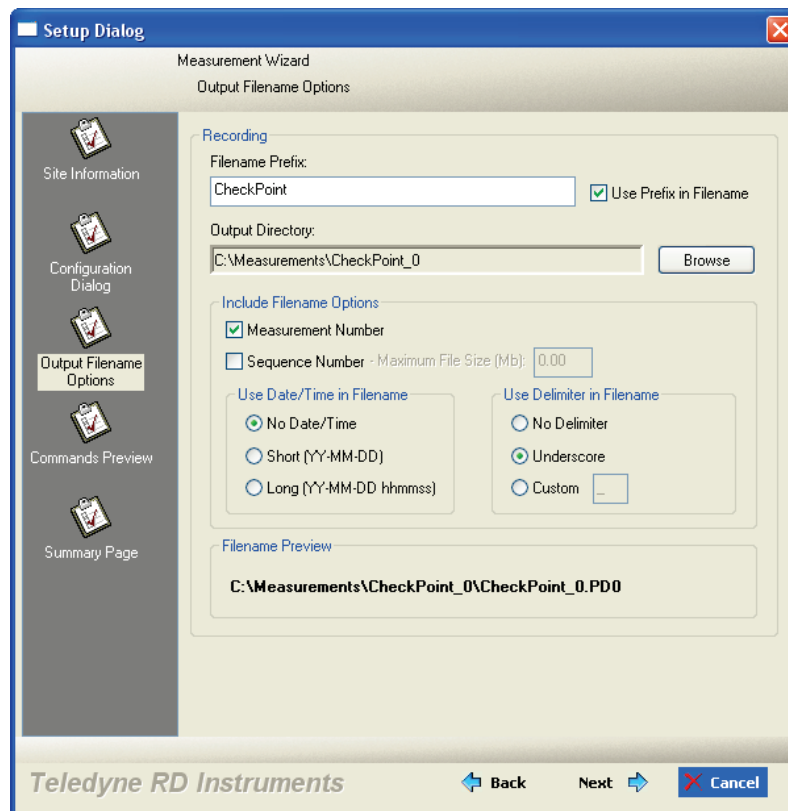


Figure 29. Output Filename Options

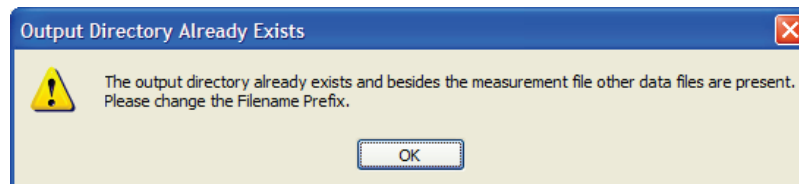
Enter your choices for the **Output Filename Options**. The **Filename Prefix** and **Output Directory** are part of the **Recording** page (see [Recording Page](#)).

Recording

Enter a **Filename Prefix** if you did not enter one on the **Site Information** page of the measurement wizard. The **Output Directory** will default to C:\Measurements\Station. Click the **Browse** button to change the output directory.



NOTE. The **Output Directory** can not already exist – You will be prompted to change the Output Directory or Filename Prefix.



Include Filename Options

- Check the **Measurement Number** box to add it to the file name.
- Check the **Sequence Number** box and enter a **Max File Size** if you want to limit the size of the data file. Once the file size has been reached, the sequence number will increment.
- To add the Date/Time to the filename, check the **Short (YY-MM-DD)** or **Long (YY-MM-DD hhmmss)** button. Select **No Data/Time** to remove the Date/Time from the filename.
- Select what type delimiter to use in the filename by selecting **No Delimiter**, **Underscore**, or **Custom**. If **Custom** is selected, enter the delimiter in the box.

Review how the filename appears in the **Filename Preview** section.

In the following example, the filename prefix is CheckPoint, the measurement number is 13, the sequence number box, measurement number, and short date/time buttons were selected.

```
C:\Measurements\CheckPoint\CheckPoint_13_000_07-04-09.PDO.
|                                     | | | |
|                                     | | | | Short Data/Time
|                                     | | | | Sequence Number
|                                     | | | | Measurement Number
|                                     | | | | Filename Prefix
|                                     | | | |
Output Directory
```

7.4 Commands Preview

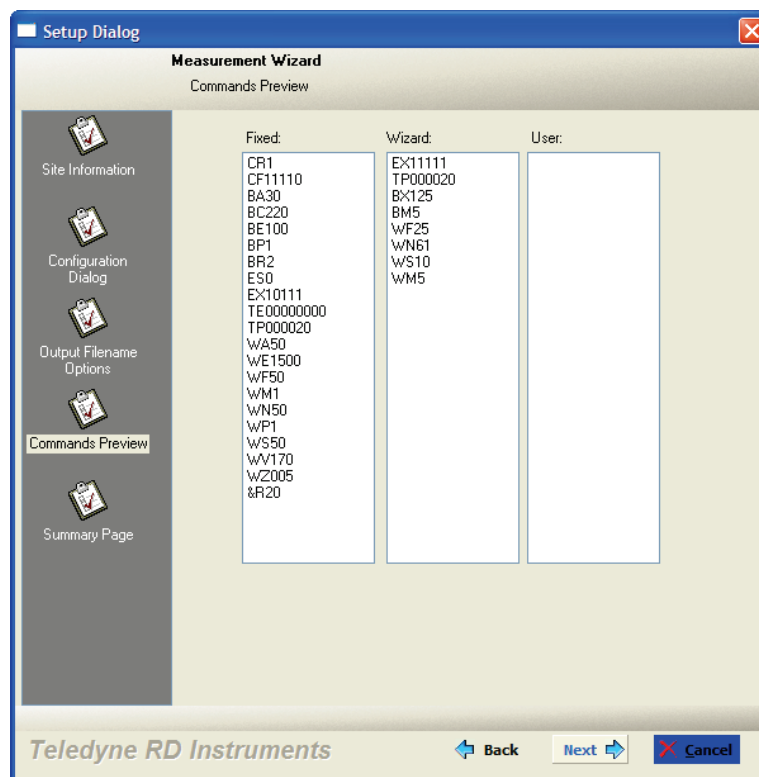


Figure 30. Commands Preview

You can directly control the profiling parameters sent to the ADCP using the **User Commands** box.



CAUTION. Do not enter any commands in the **User** section unless you are fully aware of what the command does.



NOTE. For more information on ADCP commands, see the WorkHorse Commands and Output Data Format Guide.

7.5 Summary Page

Setup Dialog
Measurement Wizard
Summary Page

Site Information

Configuration Dialog

Output Filename Options

Commands Preview

Summary Page

Devices

ADCP Type: Rio Grande 1200 Communications: COM1, 115200, 8, N, 1 ✓
Serial Number: 1704 ✓ WM12 ✓ BM7

User Supplied Information

Recording

Filename Prefix: Station1
Output Directory: C:\Measurements\Station1_0

Discharge

Flow Type is: Normal
Sample Time For Each Section [sec]: 40
Meas. Angle [deg]: 0.00

ADCP Wizard Configuration

Max. Water Depth [']: 5.00
Secondary Depth [']: 0.00
Max. Water Speed [m/s]: 0.50
Streambed: Sand
Water Mode: 11, WZ3
Bottom Mode: 5
Update Rate: 1.0
Bin Size [cm]: 5
Number of Bins: 116

Offsets

Transducer Depth [']: 0.200

Measurement Characteristics

Minimum Profile Depth [1 cell, ']: 0.57 Horizontal Velocity Std. Dev. [m/s]: ---
Minimum Profile Depth [2 cell, ']: 0.63
Maximum Profiling Depth [']: 6.69

Teledyne RD Instruments

Back Finish Cancel

Figure 31. Summary Page

Review the **Summary Page**; use the **Back** button to return to previous pages and make any necessary changes.

- If a red **X** appears next to the communication settings, it means the communications were not configured yet.
- A green **✓** next to the WM12 and BM7 means that the ADCP is capable of using these modes, not that the mode is selected.

Click the **Finish** button to complete the wizard.



NOTE. The summary does not include consequences caused by entering any **User** commands.

Measurement Characteristics

Minimum Profile Depth (1 cell) – Shows the minimum profiling depth for one cell. This depth includes the blanking distance and transducer depth.

Minimum Profile Depth (2 cells) – Shows the minimum profiling depth for two cells. This depth includes the size of bins 1 and 2, blanking distance, and the transducer depth.

Maximum Profile Depth (meters) – Shows the maximum profiling depth for the settings used.

Horizontal Velocity Standard Deviation (meters/second) – Not implemented at this time.



NOTE. Coefficient of Variation for *Section-By-Section* is defined as (Standard Deviation of the average velocity for each ping)/(average velocity of all pings).

Where the average velocity is the average of all good cells in the profile.

8 Configuration Node Properties

This section has detailed explanations of each of the settings contained in the configuration node. There are four pages that configure different portions of the configuration node.

To view the field or playback configuration node settings, right-click on the configuration node on the **Measurement Control** window, and then select **Properties** (see [Using the Measurement Control Window](#)). The configuration node presently in use will be in bold text.



NOTE. Changes to the **Field Configuration** node can only be made when not pinging. If you make any changes, you must change the **Filename Prefix** or **Output Directory** to create a new measurement file.

When creating configuration nodes for data collection, the most important parts in a configuration node are the **Commands** and **Recording** tabs. Set them to the desired parameters. The rest of the parameters can be changed during playback and do not influence data collection.

8.1 Commands Page

You can directly control the profiling parameters sent to the ADCP using the **User Commands** box. The ADCP Technical Manual explains all direct commands in detail. The **Fixed Commands** box lets you view the direct commands that will always be sent to the ADCP. The Measurement Wizard will enter the **Wizard Commands** based on information entered in the Measurement Wizard (see [Using the Measurement Wizard](#)).



CAUTION. The fixed commands are sent before any user commands. Sending a **User Command** will **OVERRIDE** the **Fixed** and **Wizard Commands**.

When the data collection is first started, the commands in the **Fixed Commands** box are sent by *Section-By-Section* to the ADCP to set its profiling parameters. The **Wizard Commands** are sent next and will *override* some

of the **Fixed Commands**. Refer to the WorkHorse Commands and Output Data Format guide for more detailed information about each command.

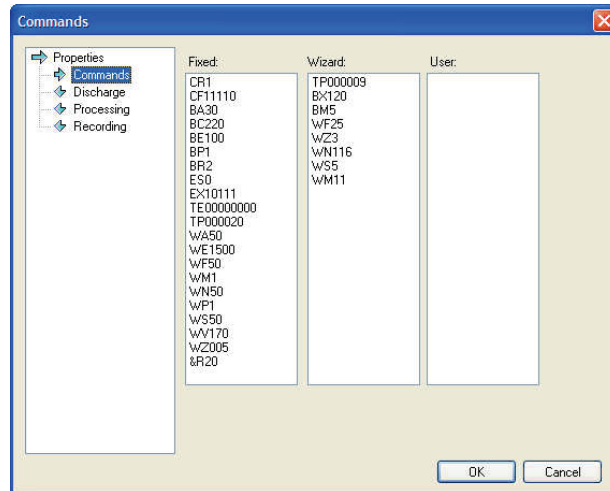


Figure 32. Commands Page

Table 2: Fixed Commands

Command	Choices	Description
CR1	Sets factory defaults	This is the first command sent to the ADCP to place it in a "known" state.
CF11110	Flow control	CF11110 selects automatic ensemble cycling, automatic ping cycling, binary data output, enables serial output, and disables data recording.
BA30	Evaluation Amplitude Minimum	Sets the minimum value for valid bottom detection to 30 counts.
BC220	Correlation Magnitude Minimum	Sets minimum correlation magnitude for valid bottom track velocity data to 220 counts.
BE100	Bottom Track Error Velocity Maximum	Sets maximum error velocity for good bottom-track water-current data to 100mm/s.
BP1	Bottom track pings	The ADCP will ping 1 bottom track ping per ensemble.
ES0	Salinity	Salinity of water is set to 0 (freshwater).
EX10111	Coordinate transformations	Sets Ship coordinates, use tilts, allow three-beam solutions, and allow bin mapping to ON.
TE00:00:00.00	Time per ensemble	Ensemble interval is set to zero.
TP00:00.20	Time between pings	Sets the time between pings to 0.2 seconds.
WA50	False Target Threshold Maximum	Sets a false target (fish) filter to 50 counts.
WE1500	Water Track Error Velocity Threshold	Sets the maximum error velocity for good water-current data to 1500mm/s.
WF50	Blank after transmit	Moves the location of the first depth cell 50 cm away from the transducer head (see Table 3).
WM1	Water mode	Sets the ADCP to Water Track mode 1.
WN50	Number of depth cells	Number of bins is set to 50 (see Table 3).
WP1	Pings per ensemble	The ADCP will ping 1 water track ping per ensemble.
WS50	Depth cell size	Bin size is set to 0.5 meters (see Table 3).
WV170	Ambiguity velocity	Sets the maximum relative radial velocity between water-current speed and WorkHorse speed to 170 cm/s.
WZ005	Mode 5 Ambiguity Velocity	Sets the minimum radial ambiguity for profiling Mode 5 (WM5) and Mode 8 (WM8) Ambiguity Velocity to 5 cm/s.
&R20	Bottom Illumination	The &R command is used to set the "Bottom Illumination". This value determines the size of the Bottom Track transmit pulse in relation to the Depth. &R is entered in percent. If you were bottom tracking in 100m of water and had &R=20, the Bottom Track transmit pulse would be 20% of 100m, or 20m. If &R=30, then the transmit pulse would be 30m.



NOTE. The above commands are shown for example only. They are valid for a 1200 kHz Rio Grande ADCP using the default settings for the wizard.

Table 3: Wizard Commands

Command	Choices	Description
EX11111	Coordinate transformations	Sets Earth coordinates, use tilts, allow three-beam solutions, and allow bin mapping to ON.
TPxx:xx.xx	Time between pings	Sets the time between pings.
BXxxx	Maximum Bottom Search	Limits how far the ADCP will search for the bottom
BMx	Bottom Mode	Sets the Bottom Mode
WFxx	Blank after transmit	Moves the location of the first depth cell away from the transducer head.
WZx	Mode 5 Ambiguity Velocity	Sets the minimum radial ambiguity for profiling Mode 5 (WM5) and Mode 8 (WM8) Ambiguity Velocity.
WNxxx	Depth cell number	Based on the maximum water depth entered in the Configuration Wizard, Section-By-Section will set the depth cell size and number of depth cells.
WSxxx	Depth cell size	Sets the depth cell size
WMx	Water Mode	Sets the Water Mode



NOTE. The values for the **Wizard Commands** are based on information entered in the **Measurement Wizard** (see [Using the Measurement Wizard](#)).

8.2 Discharge Page

Section-By-Section uses these setting to determine what formulas and calculations will be used to determine the discharge value.

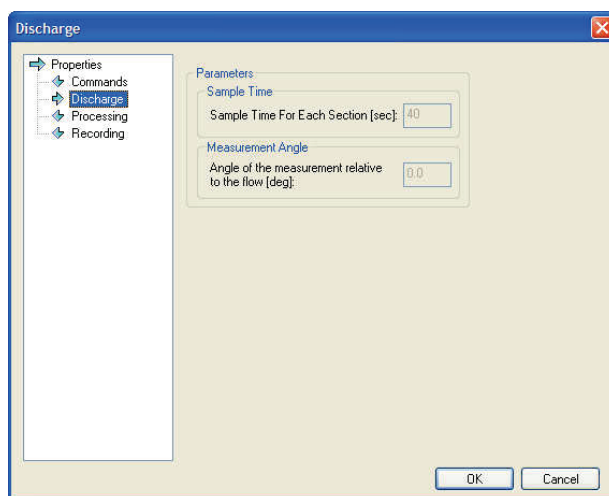


Figure 33. Discharge Page

8.3 Processing Page

The **Processing** page lets you set several system processing options and save them to a configuration node. Most values in the **Processing** tab affect the displays during data collection and post-processing. *Section-By-Section* saves these values only to the configuration node, not to the raw data files.

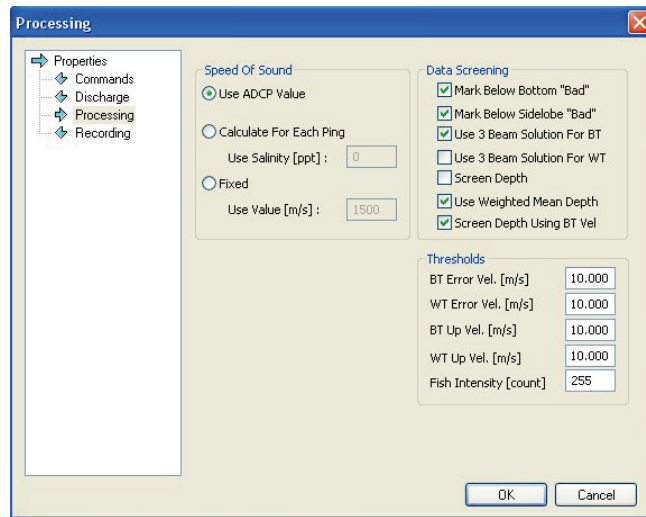


Figure 34. Processing Page

Speed of Sound

The **Speed Of Sound** section lets you correct velocity data for speed of sound variations in water. *Section-By-Section* can make these corrections dynamically with every ping or use a fixed speed of sound value.

- Use the **Use ADCP Value** option to select the value being used by the ADCP. The EC and EZ-commands determine the ADCP's speed of sound value. Choosing the **Use ADCP Value** tells *Section-By-Section* not to do any speed of sound scaling of velocity data after it is received from the ADCP.
- Selecting the **Calculate For Each Ping** option uses the **Salinity Value**, ADCP transducer depth, and the water temperature at the transducer head to compute speed of sound for each raw ADCP ensemble. *Section-By-Section* then uses this value to scale the ADCP velocity data dynamically. *Section-By-Section* uses scaled velocity data in the displays and for discharge calculations.
- Use the **Fixed** option to set a fixed value for sound speed. Select this option if you made a mistake during data collection. *Section-By-Section* uses the fixed value to re-scale the velocity data from the ADCP.

Data Screening

Select **Mark Below Bottom Bad** to mark data below the ADCP-detected bottom or Depth Sounder detected bottom (if selected for processing).

Check the **Use 3 Beam Solution for BT** box to allow 3-beam solutions if one beam is below the correlation threshold set by the BC command.

Check the **Use 3 Beam Solution for WT** box to allow 3-beam solutions if one beam is below the correlation threshold set by the WC command.

Check the **Screen Depth** box to allow for depth screening or if the ADCP sometimes reports the wrong depth. This can happen due to the detection of the reflection of the bottom. The beam is not used in the mean depth calculations if its depth is more than 75% different from the other beams.

Check the **Use Weighted Mean Depth** box to have *Section-By-Section* calculate valid depth using the following formulas.

$$d = \frac{\sum_{i=1}^n w_i d_i}{\sum_{i=1}^n w_i} \text{ and } w_i = \frac{1 - d_i}{\sum_{j=1}^n d_j}$$

Where

n = Number of valid beams

d_i = Beam depth

w_i = Weight

Check the **Screen Depth Using BT Vel** box to allow depths to be filtered out when bad bottom velocity data is detected. Use of this flag could possibly remove depth spikes from the data.

Thresholds

Bottom Track Error Velocity - The ADCP uses this parameter to determine good bottom-track velocity data. If the ADCP's error velocity value exceeds this threshold, it flags data as bad for a given depth cell.

Water Track Error Velocity - The ADCP uses this parameter to set a threshold value used to flag water-current data as good or bad. If the ADCP's error velocity value exceeds this threshold, it flags data as bad for a given depth cell.

Bottom Track Up Velocity - The ADCP uses this parameter to determine good bottom-track velocity data. If the ADCP's upward velocity value exceeds this threshold, it flags data as bad for a given depth cell.

Water Track Up Velocity - The ADCP uses this parameter to set a threshold value used to flag water-current data as good or bad. If the ADCP's upward velocity value exceeds this threshold, it flags data as bad for a given depth cell.

Fish Intensity - The ADCP uses this parameter to screen water-track data for false targets (usually fish). If the threshold value is exceeded, the ADCP rejects velocity data on a cell-by-cell basis for either the affected beam (fish detected in only one beam) or for the affected cell in all four beams (fish detected in more than one beam). Enter a value of 255 to turn off the Fish Intensity screening.

8.4 Recording Page

The **Recording** page lists the parameters used to define where the data is recorded during data collection.

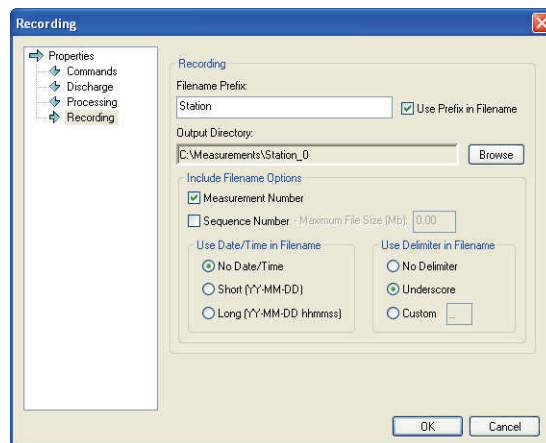


Figure 35. Recording Page

Filename Prefix – *Section-By-Section* uses the **Filename Prefix** to create the data file names made during data collection. Use the **Output Directory** field to select where the data file will be stored.

During data collection, if the **Filename Prefix** is set to TEST, *Section-By-Section* creates data files with the prefix TEST (TEST_0_000.PD0) and stores them in the directory specified in the **Output Directory** field until the disk is full. *Section-By-Section* will then stop data collection and alert you to the disk space problem.



NOTE. A file name can contain up to 255 characters, including spaces and the **Filename Prefix**. It cannot contain the following characters: \ / : * ? " < > |

Use Date/Time in Filename – Check this box if you want the date and time stamp to be added to the file name.

Maximum File Size – Select the **Sequence Number** box to use the **Maximum File Size** field to limit the size of a data file. The default for the maximum data file size is unlimited (**Sequence Number** box is not selected). If you set the Maximum File Size to 1.44, then when the size of the recorded data file reaches 1.44 MB, the **Sequence Number** increments.

9 QA/QC

Before taking measurements, check the following items.

9.1 Set the ADCP Clock

- On the **Acquire** menu, click **Set ADCP Clock**.
- Click the **Set Clock** button to set the ADCP's time to the GPS time (if available) or the PC's time. If necessary, set the PC's clock first.
- Click **Close** to exit the **Set ADCP Clock** dialog.



NOTE. The first time a transect is started, the **Set ADCP Clock** dialog will open. StreamPro ADCP users will notice that the blue LED will blink until the **Close** button is clicked. This is normal.

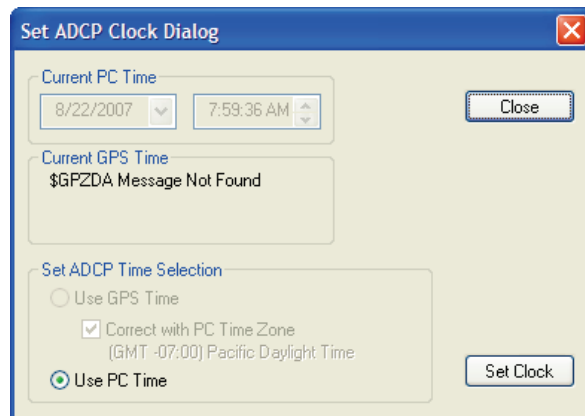


Figure 36. Set the ADCP Clock

9.2 Test the Pressure Sensor

The pressure sensor test should be done with the ADCP out of the water.

- On the **Acquire** menu, click **Execute Pressure Sensor Test** to verify the ADCP's pressure sensor is functioning properly.
- Click the **Zero Pressure Sensor** button to zero out the sensor.
- Click **Close** to exit the **Pressure Sensor Calibration** dialog.



NOTE. If your ADCP does not have a pressure sensor, you will see an error message.

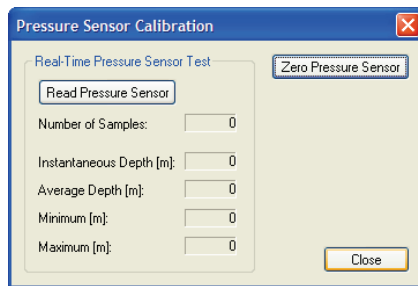


Figure 37. Test the Pressure Sensor

9.3 Test the ADCP

- On the **Acquire** menu, click **Execute ADCP Test** to verify the ADCP is functioning properly.
- Click the **Stop PC2** button to end the PC2 test. Click **Close** to exit the **ADCP Test** dialog.



NOTE. The tests should be run while the ADCP is in non moving water. Running the test in air will not harm the ADCP, but some tests may fail in air.

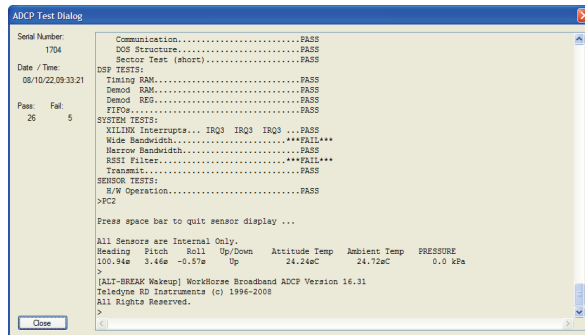


Figure 38. Test the ADCP

9.4 Compass Calibration

- On the **Acquire** menu, click **Execute Compass Calibration**.
- Click on the **Calibrate** button to begin the compass calibration.



NOTE. For more information on the Compass Calibration, see the WinRiver II User's Guide, Integrating Depth Sounder, External Heading, and GPS Data.

10 Acquiring Discharge Data

When data collection is started, *Section-By-Section* sends the direct commands from the configuration to the ADCP and controls data recording for acquisitions of individual transect discharge data. Data can be displayed in tabular, profile, or contour formats. *Section-By-Section* calculates discharge as it is calculated in real-time and accumulated as a section is made.



NOTE. It is a good idea to test the ADCP before collecting data at a particular site to check the ADCP's operation (see [QA/QC](#)).

- a. Start *Section-By-Section* and load or create a measurement file (see [Using the Measurement Wizard](#)).
- b. You may want to record a note about the instrument setup or factors such as wind conditions, the passage of other vessels, and any other noteworthy events that occur during your transect of the channel. Right-click **Site Information** on the **Measurement Control** window to add a note to the measurement file (see [Add Note](#)).
- c. Press **F4** to start the ADCP pinging. The **AcquireControl** window will now show [ADCP PINGING](#).

Section-By-Section will send the commands generated by the configuration wizard to the ADCP.

- d. Review the **Command Log** to see if any error occurs. If the command log closed before you finished reviewing the commands, click **Configure, Command Log...** Scroll through the log file and make sure no command generated an error message.

In a few seconds, you will see changing values in several of the data windows. The ensemble number will update, the time displayed just below the ensemble number will change to show the ADCP time of each ensemble, the position sensors will update, and so on. And, if you have the ADCP in sufficiently deep water, you will see velocity profiles on the graph. *Section-By-Section* will not display meaningful data if your system is not in water.

In general, after you have started the ADCP pinging, there is no reason to stop pinging until you are finished and ready to remove the ADCP from the water. An exception would be if you were operating off of batteries with limited capacity. Then, stop pinging to conserve power.



NOTE. You can let the ADCP ping even if it is not in the water. Unlike many other acoustic devices, no damage to the ADCP will occur. It is often useful to operate the ADCP out of the water to test the configuration, practice with *Section-By-Section*, or perform other tests.

- e. Before going forward, on the **Configure** menu, click **Reference**. Choose **None** from the list. *Section-By-Section* is assuming that the ADCP is not moving. You can always determine what velocity reference is in use by looking at the text lines just above and below the scales at the top and bottom of the graph display. At the end of each line is an indication of what reference is being used.

10.1 Measurements

Best practices encourage collecting 20 or more sections. Each section should be sized so that it has less than 5% of the total rated discharge Q . The following sequence must be repeated for each measurement.

10.1.1 Starting Edge

- a. Move the ADCP to the start point of the measurement.
- b. Click the **Acquire** menu and select **Start Measurement**. The shortcut key is **F5**.
- c. Since this is the first measurement, it will be at an edge. Select the edge you will start from (**Left Edge** or **Right Edge**) and enter the position of the edge in the **Edge of Water** box.
- d. Click **OK** to continue.

Figure 39. Start Measurement Dialog

10.1.2 Sections

- a. Now move to the first vertical to be measured.
- b. Enter the **ADCP Depth (m)**. This value will be used for this (and following) sections until changed.

- c. If the measurement line was not perpendicular to the flow, then enter the angle of deviation from perpendicular in the **Section Flow Angle** box (see [Measurement Angle](#)). This needs to be done for each section.
- d. Enter the **Flow Type**. If the velocity profile is expected to be non-classical, wind driven at the surface for example, then selecting the **Bi-directional** box will use 3-point extrapolation at the surface and no-slip extrapolation to the bottom.
- e. Enter the position of the **Current Station**.
- f. Click **OK** to continue.

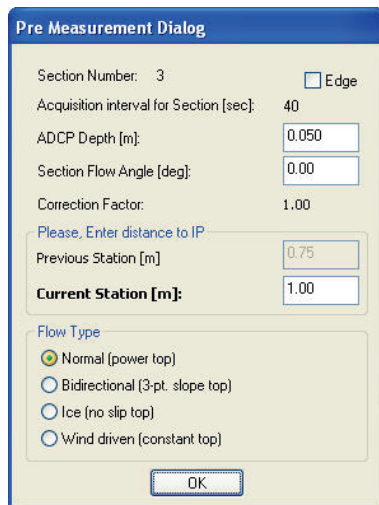
The image shows a software dialog box titled "Pre Measurement Dialog". It contains several input fields and a section for flow type selection. The fields are: "Section Number" with a value of 3, "Acquisition interval for Section [sec]" with a value of 40, "ADCP Depth [m]" with a value of 0.050, "Section Flow Angle [deg]" with a value of 0.00, "Correction Factor" with a value of 1.00, "Previous Station [m]" with a value of 0.75, and "Current Station [m]" with a value of 1.00. There is a checkbox labeled "Edge" which is currently unchecked. Below the input fields is a section titled "Flow Type" with four radio button options: "Normal (power top)" (which is selected), "Bidirectional (3-pt. slope top)", "Ice (no slip top)", and "Wind driven (constant top)". An "OK" button is located at the bottom right of the dialog box.

Figure 40. Pre Measurement Dialog

- g. Now the data will be recorded. The **Duration** near the center of the **Composite Tabular 1** screen will count up to the time remaining in the measurement interval (default 40 seconds).

As the ADCP pings, the **Average Velocity** of all of the pings is displayed as well as the **Coefficient of Variation** of all of the pings to that time. The Coefficient of Variation should slowly decrease and reach a relatively stable value. The depth of the water is also displayed along with the number of seconds that have elapsed since pinging began.

- h. When the measurement interval has expired, the **Post Measurement Dialog** will appear giving a summary of the data for that station. The discharge calculated for that panel is shown followed by the percentage of the rated Q that is measured at this station.

If the discharge for the station is more than 5% of the rated Q, these values will be show in red text; otherwise, they will be shown in black text.

- i. If you are satisfied with the results for this station, click **Accept Data**. If you want to repeat the station or if you want to change the position before repeating, click the **Reject and Repeat** button.

Post Measurement Dialog

Recorded Data for Measurement: 3

Calculated Discharge [m ³ /s]:	0.096
Calculated Discharge % Rated:	0.0
Average Velocity [m/s]:	0.536
Coefficient of Variation [%]:	12.3
Average Depth [m]:	0.718
Number of Valid Pings:	40
Section Flow Angle CF:	1.00
Standard Deviation Flow Direction:	6.33

Past Discharge Data

Previous Section Discharge [m ³ /s]:	0.071
Accumulated Section Discharge [m ³ /s]:	0.167

OK Cancel

Figure 41. Post Measurement Dialog

- j. Move to the next station.
- k. Press **F5** to start the measurement.
- l. Wait for the data to be collected.
- m. If you are satisfied with the results for this station, click **Accept Data**.
- n. Repeat steps “g” through “j” and collect 20 sections.

10.1.3 Ending Edge

- a. Check the **Edge** box when you reach the last measurement position.

Pre Measurement Dialog

Section Number: 21 ☒ Edge

Acquisition interval for Section [sec]: 40

Section Flow Angle [deg]: 0.00

Confirmation Dialog:

? You have indicated that you have completed the vertical measurements and are ready to proceed to the edge of water measurement. Do you wish to continue?

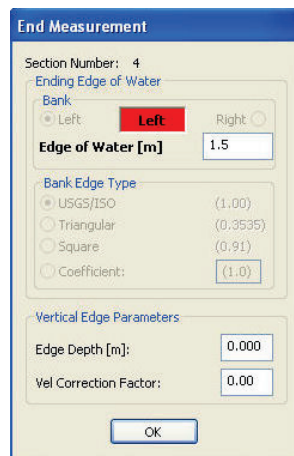
Yes No

OK

Figure 42. Edge Selected

- b. Click **Yes** to finish the measurement.

- c. The selected edge will be the opposite edge from the one you selected at the beginning.
- d. Enter the position of the final edge. If this edge is a vertical wall, then also enter the depth and the appropriate correction factor.
- e. Click **OK**.



The image shows a software dialog box titled "End Measurement". It contains several input fields and radio buttons. At the top, "Section Number" is set to 4. Below that, "Ending Edge of Water" has a "Bank" section with "Left" selected (indicated by a red background) and "Right" unselected. Next to it is a text box for "Edge of Water [m]" containing the value 1.5. Below this is a "Bank Edge Type" section with four options: "USGS/ISO" (selected), "Triangular", "Square", and "Coefficient:". Each option has a corresponding value in parentheses: (1.00), (0.3535), (0.91), and (1.0) respectively. At the bottom is a "Vertical Edge Parameters" section with "Edge Depth [m]" set to 0.000 and "Vel Correction Factor" set to 0.00. An "OK" button is at the very bottom.

Section Number	Ending Edge of Water	Edge of Water [m]	Bank Edge Type	Edge Depth [m]	Vel Correction Factor
4	Left	1.5	USGS/ISO	0.000	0.00

Figure 43. End Measurement Dialog

10.2 Data Collections Tips

- Check the **Edge** box (**Left** or **Right Edge**) for the first and last sections and enter the distance of the edge of the water. If the edge is vertical instead of sloped, enter the **Edge Depth** and enter the **Correction** factor.
- Minimize the ADCP movement while section measurements are recorded.
- Collect 20 or more sections.
- TRDI suggests doing a quick survey of the stream using the *WinRiver II* software to get a value for the rated Q, if it is not known for the site.
- Each section should be sized so that it has less than 5% of the total rated discharge Q.
- Station distances can count up (1.5 meters, next station 2.0 meters) or down (2.0 meters, next station 1.5 meters) as necessary.

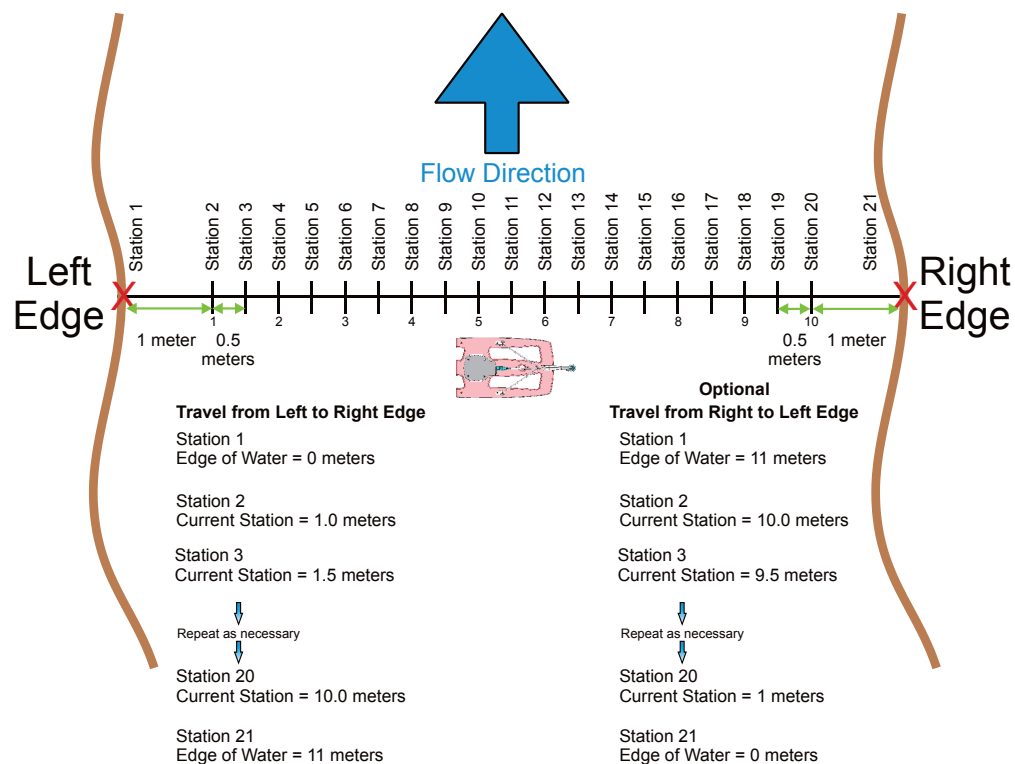


Figure 44. Overview of Data Collection

11 Post-Processing of Discharge Data

Section-By-Section is used for post-processing data to get a total discharge value for the channel (see [Using the Q Measurement Summary](#)). After collecting 20 sections on the water body, each station must be verified to be within 5% of the mean discharge calculated for the set. If any of the stations are outside of the tolerance, additional measurements should be taken.

11.1 Playback a Data File

- Start *Section-By-Section*.
- On the **File** menu click **Open Measurement...**
- Select the measurement file (*.mmt) to be played and click **Open**.
- To playback a measurement file, right-click on the measurement file and select **Reprocess Measurement**. This will create a copy of the **Field Configuration** node called **Playback Configuration**. The data files will automatically play to the end of the file.

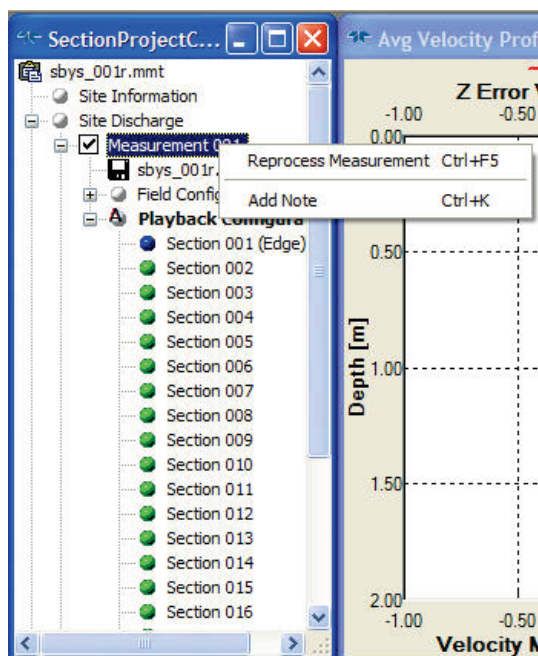


Figure 45. Reprocess Measurement

11.2 Playback Older Data Files

Use the following procedure to playback a data file created using the StreamPro version of *Section-By-Section*.

- a. Start *Section-By-Section*. On the **File** menu click **Create Measurement from Data Files** and select a data file to be played. *Section-By-Section* automatically reads the *.xml configuration file and create the **Field Configuration** node.
- b. To open a workspace file, click **File, Load Workspace**.
- c. On the **Measurement Control** window, right-click on the measurement file and select **Reprocess Measurement**. The playback tool bar has functions to start, stop, rewind, and go to the end of the data file.
- d. Use the **Measurement Control** window to see what files are in use (see [Using the Measurement Control Window](#)).
- e. Play back the individual data files with their associated configuration nodes. After playing through the data file, press **View, Section Discharge Summary** or by using the shortcut key **F12** to obtain the total discharge for each measurement.

The section discharge summary shows all recent sections that have been made and all relevant information on them. It can be used after data collection and during playback and is very effective at establishing whether a measurement is good or not.



NOTE. As a standard for all ADCP users it is widely accepted that sections should be within 5% of each other assuming constant stage.

11.3 Playback Data Display Options

The Playback data display options are the same as those for data collection.

- a. On the **Playback** menu, click **First Ensemble** to go to the beginning of the data file. Click **Play** to review the data. To quickly play through the data, on the **Playback** menu, select **Slider** or drag the ensemble marker on the contour graph.
- b. Right-click the display and select **Properties** to change the scaling fields of the different data displays.
- c. Use the toolbar to zoom in on (see [Zoom Functions](#)).

11.4 Editing an Item During Playback

Some of the parameters in the **Playback Configuration** can be changed while playing back data. The parameters that controlled data collection (**Recording** and **Commands** pages) can not be changed.



NOTE. To return the all values to the parameters as collected, on the **Measurement Control** window, right-click on **Playback Configuration** and select **Reset Properties**.

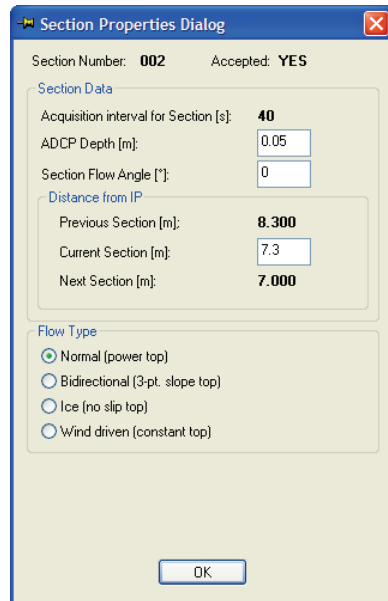
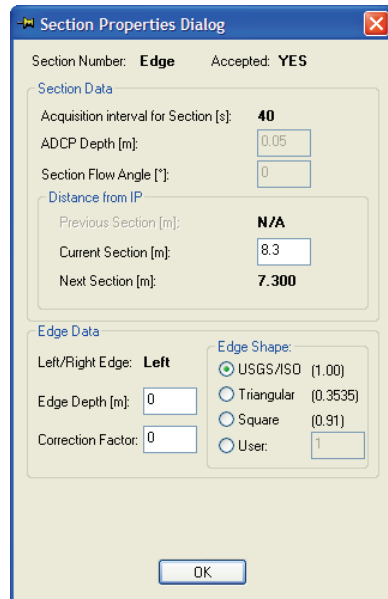


Figure 46. Section Properties

Change Section Properties

- Open the measurement file.
- On the **Measurement Control** window, right-click on a **section** node and select **Properties**.
- Enter the corrections.
- Click **OK**.

Change Edge Properties



The image shows a 'Section Properties Dialog' window. At the top, it says 'Section Number: Edge' and 'Accepted: YES'. Below this is a 'Section Data' section with fields for 'Acquisition interval for Section [s]: 40', 'ADCP Depth [m]: 0.05', and 'Section Flow Angle [°]: 0'. Underneath is a 'Distance from IP' section with 'Previous Section [m]: N/A', 'Current Section [m]: 8.3', and 'Next Section [m]: 7.300'. The 'Edge Data' section at the bottom has 'Left/Right Edge: Left', 'Edge Depth [m]: 0', and 'Correction Factor: 0'. To the right of these are 'Edge Shape' options: 'USGS/ISO (1.00)' (selected with a radio button), 'Triangular (0.3535)', 'Square (0.91)', and 'User: 1'. An 'OK' button is at the bottom center.

Section Number	Accepted
Edge	YES

Section Data

Field	Value
Acquisition interval for Section [s]	40
ADCP Depth [m]	0.05
Section Flow Angle [°]	0

Distance from IP

Field	Value
Previous Section [m]	N/A
Current Section [m]	8.3
Next Section [m]	7.300

Edge Data

Field	Value
Left/Right Edge	Left
Edge Depth [m]	0
Correction Factor	0

Edge Shape

Shape	Value
USGS/ISO	1.00
Triangular	0.3535
Square	0.91
User	1

OK

Figure 47. Edge Section Properties

- Open the measurement file.
- On the **Measurement Control** window, right-click on the **edge** node and select **Properties**.
- Enter the corrections.
- Click **OK**.

11.5 Print a Graph or Display

- a. Start *Section-By-Section* and open a measurement file.
- b. Click on the graph/display to be printed. The title bar will be highlighted.
- c. On the **File** menu, click **Print Setup**. Select the desired setting and printer.



NOTE. Choose **Landscape** for contour graphs.

- d. On the **File** menu click **Print Preview**. If the display is acceptable, click **Print**.

11.6 Capture Setup

If you would like to save a graph/display for use in other documents, do the following.

- a. Click on the graph/display to be saved. The title bar will be highlighted.
- b. On the **Configure** menu, select **Capture Setup**. This will bring up a following dialog box.

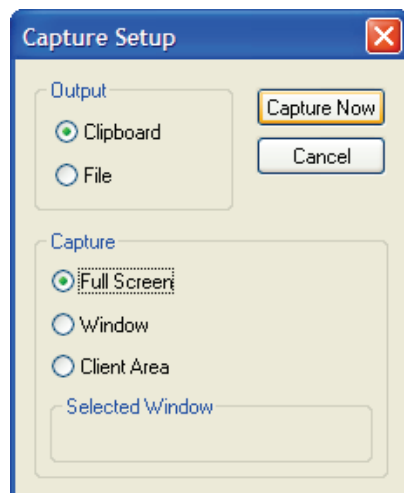


Figure 48. Capture Setup

- c. Select **Clipboard** or **File**.
- d. Select **Full Screen**, **Window**, or **Client Area**. The **Client Area** will include all parts of the graph/plot except the title bars.
- e. Click **Capture Now**. If you selected **File**, name the file and click **Save**.

11.7 Export Data

The **Section Discharge Summary** data can be exported.

- Right-click the **Section Discharge Summary** display and select **Export to CSV** to export a comma delimited file able to be used in spreadsheets and other programs.
- Select **Copy** to copy the text to the Windows® clipboard as tabular delimited text.
- Use the **Properties** function to change the type font used for the text.

Measurement	Q [m³/s]	% Total Q	Accum Q [m³/s]	% Diff.
Measurement ...	3.149			4.9
Section 001	0.000	0.00%	0.000	
Section 002	0.135	4.28%	0.135	
Section 003	0.090	2.85%	0.224	
Section 004	0.113	3.59%	0.337	
Section 005	0.142	4.50%	0.479	
Section 006	0.164	5.19%	0.643	
Section 007	0.187			
Section 008	0.194			
Section 009	0.187			
Section 010	0.182			
Section 011	0.177	5.64%	1.370	
Section 012	0.180	5.71%	1.750	
Section 013	0.180	5.72%	1.930	
Section 014	0.184	5.85%	2.114	
Section 015	0.188	5.97%	2.303	
Section 016	0.174	5.52%	2.476	
Section 017	0.171	5.44%	2.648	
Section 018	0.148	4.72%	2.796	
Section 019	0.130	4.12%	2.926	
Section 020	0.093	2.95%	3.019	
Section 021	0.077	2.44%	3.095	
Section 022	0.054	1.70%	3.149	
Section 023	0.000	0.00%	3.149	

Figure 49. Section Discharge Summary

11.8 Using the Q Measurement Summary

The Q Measurement Summary creates a summary of the measurement that can be printed.

- Start *Section-By-Section* and open a measurement file.
- Playback / Reprocess the measurement (see [Reprocess Measurement](#)).
- Check/edit the Site Information by right-clicking on **Site Information** in the Section Project Control window and selecting **Site Wizard**.

- d. On the **File** menu, click **Print Preview Q Measurement Summary**.
- e. On the **Q Measurements** screen, click **Print** to print a copy of the discharge summary.

Teledyne RD Instruments
Section-By-Section Discharge Summary Date: 01/07/2005

Site Information		Measurement Information	
Site Name:	chk13	Units of Measure:	SI
		Measure No.:	0
Summary:			
Width:	7.438	Area:	7.604
Velocity:	0.255	Stage:	1.000
Discharge:	1.938		
Personnel & Equipment			
Party:	me		
Rating Information			
Est. Uncert.:	3.492	L. Edge Type:	USGS/ISO
Num. Stations:	22	R. Edge Type:	USGS/ISO
Rated Q:	0.440	L. Edge Coeff.:	1.000
		Begin Shore:	RIGHT
System Information		System Setup	
Serial Number:	208	ADCP Depth:	0.050
Firmware Version:	31.03	Meas. Angle:	0.000
Software Version:	2.02.18.00	Cell Size:	0.07
System Frequency:	2.0 MHz	Num. Cells:	20

Stn #	Distance (m)	Depth (m)	Width (m)	Start Time	Meas. Time (s)	Percent Good	Coeff. of Variation	Average Velocity (m/s)	Adjusted (m/s)	Angular Corr. Factor	Area (m²)	Discharge (m³/s)	% of Total
1	0.030	0.000	0.366	12:59:05	0.00	97.44	0.000	0.000	0.000	1.000	0.000	0.000	0.00
2	0.762	0.548	0.518	12:59:05	38.00	97.44	80.126	0.059	0.059	1.000	0.283	0.016	0.81
3	1.067	0.764	0.305	12:59:43	40.00	87.80	43.693	0.099	0.099	1.000	0.233	0.023	1.19
4	1.372	0.962	0.304	13:02:04	41.00	90.48	28.743	0.154	0.154	1.000	0.293	0.045	2.32
5	1.678	1.159	0.304	13:03:20	40.00	95.12	25.359	0.161	0.161	1.000	0.353	0.057	2.94
6	1.981	1.271	0.305	13:04:44	40.00	95.12	22.890	0.196	0.196	1.000	0.388	0.072	3.69
7	2.286	1.360	0.305	13:05:42	40.00	95.12	20.565	0.193	0.193	1.000	0.415	0.079	4.09
8	2.591	1.415	0.305	13:06:40	40.00	95.12	18.215	0.209	0.209	1.000	0.431	0.090	4.65
9	2.896	1.423	0.304	13:07:37	40.00	95.12	11.964	0.239	0.239	1.000	0.433	0.104	5.35
10	3.200	1.422	0.304	13:09:04	40.00	92.68	17.191	0.263	0.263	1.000	0.433	0.114	5.88
11	3.505	1.425	0.305	13:10:10	40.00	95.12	13.990	0.277	0.277	1.000	0.434	0.120	6.21
12	3.810	1.432	0.305	13:11:13	40.00	90.24	10.757	0.291	0.291	1.000	0.437	0.127	6.56
13	4.115	1.444	0.305	13:12:15	40.00	95.12	11.662	0.290	0.290	1.000	0.440	0.128	6.61
14	4.420	1.447	0.304	13:13:47	40.00	92.68	10.123	0.310	0.310	1.000	0.441	0.137	7.05
15	4.724	1.442	0.304	13:14:51	40.00	92.68	12.244	0.327	0.327	1.000	0.439	0.143	7.40
16	5.029	1.406	0.305	13:15:48	39.00	95.00	9.863	0.320	0.320	1.000	0.429	0.136	7.03
17	5.334	1.311	0.305	13:16:47	39.00	92.50	11.480	0.333	0.333	1.000	0.400	0.134	6.90
18	5.639	1.224	0.305	13:17:47	40.00	95.12	10.042	0.321	0.321	1.000	0.373	0.121	6.24
19	5.944	1.038	0.304	13:18:57	40.00	95.12	15.301	0.319	0.319	1.000	0.316	0.103	5.34
20	6.248	0.826	0.304	13:19:56	40.00	95.12	15.862	0.321	0.321	1.000	0.251	0.081	4.17
21	6.553	0.624	0.610	13:21:23	39.00	95.00	20.735	0.281	0.281	1.000	0.381	0.108	5.56
22	7.468	0.000	0.457	13:22:02	0.00	95.00	0.000	0.000	0.000	1.000	0.000	0.000	0.00

Diagnostic Files	
Raw File: sxs_000r.000	Config. File: sxs_000r.mmt
Field Notes:	
[01/07/2005 12:55 AM] Transect 0 aborted due to inability to program instrument.	

Figure 50. Q Measurement Summary

12 Shortcut Keys

Use the shortcut keys to quickly perform tasks.

Table 4: Section-By-Section Shortcut Keys

Key	Description
F1	Help
F4	Start/Stop Pinging
F5	Start/Stop Measurement
F7	Properties
F8	Toggle Bank
F9	Toggle Ensemble Header Tabular view
F11	Toggle Detailed Discharge/Composite Tabular view
F12	Toggle Discharge Summary Tabular view
Ctrl-A	Set as Active Configuration (when Measurement Control window selected)
Ctrl-B	Reference - Bottom Track
Ctrl-C	Copy
Ctrl-D	Toggle Acquire Control window
Ctrl-E	Close Measurement File
Ctrl-F	Create Measurement from Data Files
Ctrl-F2	Apply to All Active Configurations
Ctrl-F3	Toggle Measurement Management window (when selected)
Ctrl-F5	Reprocess Measurement
Ctrl-F7	Properties (when window or file selected)
Ctrl-F8	Data Selection (when Profile or Contour plot selected)
Ctrl-U	Duplicate Configuration node (when Measurement Control window selected)
Ctrl-T	Delete Configuration node (when Measurement Control window selected)
Ctrl-T	Export to CVS (When Discharge Summary selected)
Ctrl-J	Rename Configuration node (when Measurement Control window selected)
Ctrl-R	Reset Configuration node Properties (when Measurement Control window selected)
Ctrl-I	Site Wizard
Ctrl-K	Add Note (when Measurement Control window selected)
Ctrl-L	Lock/Unlock Measurement File
Ctrl-M	View Command Log
Ctrl-N	Reference - None
Ctrl-O	Open measurement file
Ctrl-P	Print
Ctrl-S	Save Measurement File
Ctrl-W	New Measurement
Shift-F4	Set ADCP Clock
Shift-F8	Execute RGTest
Shift-F9	Execute Pressure Sensor Test
Shift-F10	Execute Compass Calibration
Minus	Previous Ensemble (Playback)
Space	Next Ensemble (Playback)
Ctrl-Minus	Previous Section (Playback)
Ctrl-Space	Next Section (Playback)
Home	First Ensemble (Playback)
End	Last Ensemble (Playback)

13 Software History

Version 2.07

- Fixed missing "Configure" menu item in foreign language versions.

Version 2.06

- Removed Add/Remove Option since Section by Section option is now stored in Firmware.
- Added Compass Compatibility for new StreamPro Firmware.
- Fixed several RiverRay crashing issues.
- Added support for enhanced 6m StreamPro Long Range mode.

Version 2.05

- Fixed problem with the calculation of top discharge for no-slip method (ice measurement).
- Fixed occasional crashing problem when reading xml nodes.
- Fixed problem with velocities in the Measurement Summary being wrong for Ice and Bidirectional methods.
- Fixed problem with saving measurement file in playback.
- Fixed problem with ADCP depths entered in English units.
- Added a check to prevent illegal characters in file and folder names.
- Fixed problem in acquire with sections occasionally being duplicated.
- Fixed problem with Section Summary when no ensembles were recorded for the section.
- Fixed crash when closing section's property window during acquire.
- Changed WT Error Velocity Thresholds for StreamPro to depend on predicted standard deviation.
- Fixed problem with wizard generating bad commands if StreamPro was chosen but not connected to Com Port while running the wizard.
- Fixed Wizard Auto Mode Selection to be the same as in WinRiver II, Moving Boat.
- Fixed problem with WT Error Velocity not being set for Rio-Grande. It now matches Moving Boat.

Version 2.04

- Add a flag to Processing page to turn on/off screening of bottom depths based on bottom velocity.

Version 2.03

- Added ability to set ADCP depth for each section.
- Added ability to edit section location in playback.
- Added ability to mark a section to not be used in playback.
- Added ability to use the screening thresholds.
- Added new velocity magnitude graph for viewing/evaluating the extrapolation profile for each section.
- Added ability to set extrapolation on a per section basis (ice, bidirectional, normal, wind).
- Added ability to reload a *.mmt file created earlier and allow data collection to commence.
- Added ability to playback the portion of a river measured up to the point of a crash.
- Fixed problem with printing summary when files are on a network.
- Removed 'StreamPro' from the title in Q measurement summary.
- Disabled selection of Set ADCP clock and other tests while pinging.
- Improved behavior when pressing Ctrl + F5 too many times in a row.
- Fixed some problems when importing files collected with StreamPro software.
- Updated language pack.

Version 2.02

- Changed rounding of StreamPro maximum depth when in English units.
- Reset section number if pinging stopped.
- Changed logic for StreamPro Mode 13 valid pings calculation to remove error message.
- Changed measurement time to zero for the edges.
- Changed end edge start time to account for the time the previous section lasted.
- Changed measured discharge calculation.
- Fixed flag for Weighted mean depth value when exporting StreamPro xml file.

- Changed default for EX-command for Rio Grande from earth to ship.

Version 2.01

- Added capability to reprocess with different section angles than the data was collected.
- Changed layout of Composite Tabular display.
- Changed behavior after re-processing so the program re-positions to ensemble/section selected by the user.
- Fixed problem with calculating mean velocity during acquire after changing IP Distance.
- Fixed problem with discharge not being updated after changing velocity reference.
- Fixed value of "Distance from IP" in CSV file.
- Fixed value of "Total_Width" in CSV file when saving file in English units.
- Fixed problem with "Section-By-Section Report" not being updated with some values from Site Information page.
- Changed resolution of "Correction Factor" on "Section Properties" dialog.
- Removed "Transect Number" from the wizard.
- Fixed problem with decoding some values from StreamPro XML file.
- Changed auto-scaling of "Section Contour" graph so section and ensemble velocity are scaled the same.
- Fixed "Bottom Q" value on section contour graph.
- On the "Section Properties" dialog added display of the "Edge Shape".
- Added pre-filling of some Acquire Wizard fields.
- Fixed problem updating Composite Tabular when Flow Type was changed.
- Fixed problem handling poor bi-directional flow type data set.
- Fixed problem in Section Properties Dialog: Acquisition interval was always 40 sec.
- Fixed problem with bad distance value in "Section Discharge Summary" if section was rejected.

Version 2.00

- Initial release.

NOTES

NOTES